

Machine Learning Laboratory

Experiment:

Acute Ischemic Stroke Risk Classification Engine – Dataset Exploration & Exploratory Data Analysis

Dataset:

Healthcare Stroke Prediction Dataset

Question 1

Aim

Import, load and view the dataset.

Program

```
from pathlib import Path
import sys

import pandas as pd
from IPython.display import display

PROJECT_ROOT = Path.cwd()
if not (PROJECT_ROOT / 'Stroke prediction dataset').exists():
    PROJECT_ROOT = PROJECT_ROOT.parent

if str(PROJECT_ROOT) not in sys.path:
    sys.path.insert(0, str(PROJECT_ROOT))

from src.eda_utils import (
    create_bivariate_plots,
    create_categorical_countplots,
    create_correlation_heatmap,
    create_distribution_analysis_plots,
    create_missing_values_plot,
    create_numeric_distribution_plots,
    create_outlier_plots,
    create_target_plots,
    dataframe_profile,
    dataset_quality_notes,
    ensure_output_directories,
    load_dataset,
    missing_summary,
    normality_summary,
    outlier_summary,
    statistical_summary,
    target_distribution,
    target_percentages,
)

ensure_output_directories()
pd.set_option('display.max_columns', None)
pd.set_option('display.width', 120)

df = load_dataset(PROJECT_ROOT)

print('First 5 rows:')
display(df.head())

print('Last 5 rows:')
```

```
display(df.tail())

print('Dataset shape:', df.shape)
print('Column names:', list(df.columns))
print('Data types:')
display(df.dtypes.to_frame(name='dtype'))
print(f'Memory usage: {df.memory_usage(deep=True).sum() / (1024 ** 2):.2f} MB')
```

Output

Question 1 - head()

	id	gender	age	hypertension	heart_disease	ever_married	work_type	Residence_type	avg_glucose_level	bmi
0	9046	Male	67.0	0	1	Yes	Private	Urban	228.69	36.6
1	51676	Female	61.0	0	0	Yes	Self-employed	Rural	202.21	NaN
2	31112	Male	80.0	0	1	Yes	Private	Rural	105.92	32.5
3	60182	Female	49.0	0	0	Yes	Private	Urban	171.23	34.4
4	1665	Female	79.0	1	0	Yes	Self-employed	Rural	174.12	24.0

Figure 1. First five rows of the dataset.

Question 1 - tail()

	id	gender	age	hypertension	heart_disease	ever_married	work_type	Residence_type	avg_glucose_level	bmi
5105	18234	Female	80.0	1	0	Yes	Private	Urban	83.75	NaN
5106	44873	Female	81.0	0	0	Yes	Self-employed	Urban	125.20	40.0
5107	19723	Female	35.0	0	0	Yes	Self-employed	Rural	82.99	30.6
5108	37544	Male	51.0	0	0	Yes	Private	Rural	166.29	25.6
5109	44679	Female	44.0	0	0	Yes	Govt_job	Urban	85.28	26.2

Figure 2. Last five rows of the dataset.

Question 1 - Shape and Columns

Shape: (5110, 12)

Columns:

['id', 'gender', 'age', 'hypertension', 'heart_disease', 'ever_married', 'work_type', 'Residence_type', 'avg_glucose_level', 'bmi', 'smoking_status', 'stroke']

Figure 3. Dataset shape and column names.

Question 1 - Data Types

```
id          int64
gender      object
age         float64
hypertension int64
heart_disease int64
ever_married object
work_type   object
Residence_type object
avg_glucose_level float64
bmi         float64
smoking_status object
stroke      int64
```

Figure 4. Data types for each column.

Question 1 - info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5110 entries, 0 to 5109
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   id                    5110 non-null   int64
1   gender                5110 non-null   object
2   age                   5110 non-null   float64
3   hypertension          5110 non-null   int64
4   heart_disease         5110 non-null   int64
5   ever_married          5110 non-null   object
6   work_type             5110 non-null   object
7   Residence_type        5110 non-null   object
8   avg_glucose_level     5110 non-null   float64
9   bmi                   4909 non-null   float64
10  smoking_status        5110 non-null   object
11  stroke                5110 non-null   int64
dtypes: float64(3), int64(4), object(5)
memory usage: 479.2+ KB
```

Figure 5. 'info()' output for the dataset.

Analysis

The dataset loads correctly with 5110 rows and 12 columns.

The file includes the expected stroke-related variables and mixed data types.

The shape, column names, and memory usage confirm that the data is small enough for manual EDA.

No loading issues appear in the notebook outputs.

Conclusion

The dataset loads successfully and is ready for EDA.

Question 2

Aim

Display dataset overview and summary statistics.

Program

```
df.info()

print('\nNumerical summary:')
display(df.describe())

print('Categorical summary:')
display(df.describe(include='object'))
```

Output

Question 2 - describe()								
	id	age	hypertension	heart_disease	avg_glucose_level	bmi	stroke	
count	5110.000	5110.000	5110.000	5110.000	5110.000	4909.000	5110.000	
mean	36517.829	43.227	0.097	0.054	106.148	28.893	0.049	
std	21161.722	22.613	0.297	0.226	45.284	7.854	0.215	
min	67.000	0.000	0.000	0.000	55.120	10.300	0.000	
25%	17741.250	25.000	0.000	0.000	77.245	23.500	0.000	
50%	36932.000	45.000	0.000	0.000	91.885	28.100	0.000	
75%	54682.000	61.000	0.000	0.000	114.090	33.100	0.000	
max	72940.000	82.000	1.000	1.000	271.740	97.600	1.000	

Figure 6. Summary statistics for numerical columns.

Question 2 - describe(include='object')

	gender	ever_married	work_type	Residence_type	smoking_status
count	5110	5110	5110	5110	5110
unique	3	2	5	2	4
top	Female	Yes	Private	Urban	never smoked
freq	2994	3353	2925	2596	1892

Figure 7. Summary statistics for categorical columns.

Question 2 - Missing Values

	missing_values	missing_percentage
bmi	201	3.93
id	0	0.00
age	0	0.00
gender	0	0.00
hypertension	0	0.00
heart_disease	0	0.00
work_type	0	0.00
ever_married	0	0.00
Residence_type	0	0.00
avg_glucose_level	0	0.00
smoking_status	0	0.00
stroke	0	0.00

Figure 8. Missing value summary table.

Question 2 - Duplicate Count

Duplicate rows: 0

Figure 9. Duplicate row count.

Analysis

The numerical summary shows that age ranges from 0.08 to 82.00.

Average glucose level is strongly right-skewed, while BMI is moderately right-skewed.

`describe(include='object')` confirms that Female, Private, Yes, Urban, and never smoked are the dominant categories.

The data types remain consistent with a structured tabular dataset.

Conclusion

The summary statistics show a realistic clinical dataset with skewness and missing BMI values.

Question 3

Aim

Analyze data quality.

Program

```
profile = dataframe_profile(df)
missing_df = missing_summary(df)

display(missing_df)
print('Total missing values:', int(profile['missing_values'].sum()))
print('Duplicate rows:', profile['duplicate_rows'])

categorical_columns = df.select_dtypes(include='object').columns.tolist()
for column in categorical_columns:
    print(f'\nUnique values for {column}:')
    display(df[column].value_counts(dropna=False).to_frame(name='count'))

print('Quality notes:')
for note in dataset_quality_notes(df):
    print('-', note)

create_missing_values_plot(df)
```

Output

Question 3 - Missing Value Table

0	missing_count	missing_percentage	id	0	0.00	gender
0.00	age	0	0.00	ever_married	0	0.00
0.00	heart_disease	0	0.00	Residence_type	0	0.00
work_type	0	0.00	bmi	201	3.93	
avg_glucose_level	0	0.00	stroke	0	0.00	
smoking_status	0	0.00				

Figure 10. Missing value table.

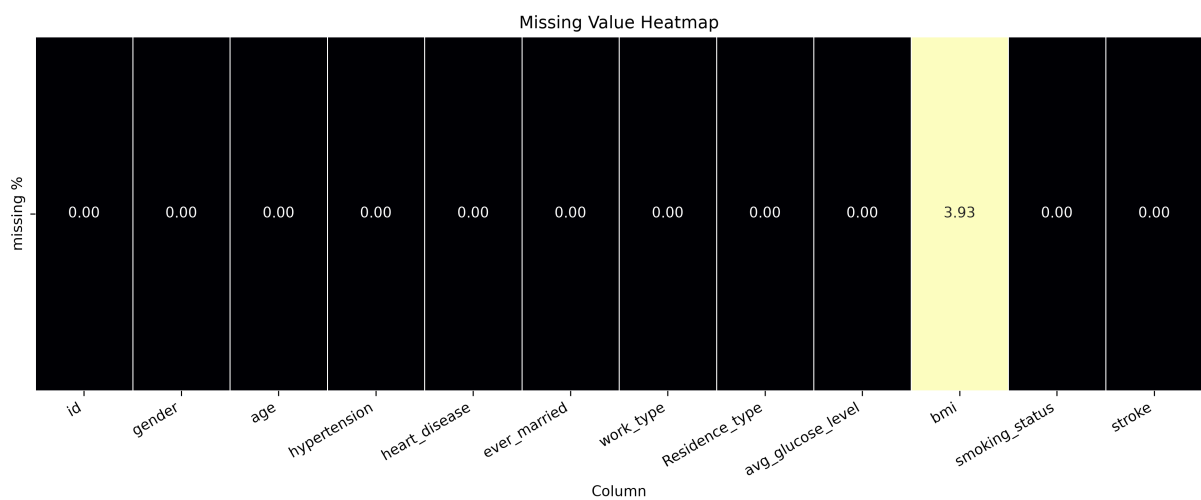


Figure 11. Missing value heatmap.

Unique values - gender					
count	gender	Female	2994	Male	2115
		Other			1

Figure 12. Unique categorical values for gender.

Unique values - work_type						
	count	work_type	Private	2925	Self-employed	819
657	Never_worked	22			children	687
					Govt_job	

Figure 13. Unique categorical values for work type.

Unique values - ever_married			
count	ever_married	Yes	3353
		No	1757

Figure 14. Unique categorical values for ever married.

Unique values - Residence_type			
count	Residence_type	Urban	2596
		Rural	2514

Figure 15. Unique categorical values for Residence type.

Unique values - smoking_status					
	count	smoking_status	never smoked	1892	Unknown
smokes	789			1544	formerly smoked
					885

Figure 16. Unique categorical values for smoking status.

Analysis

BMI is the only column with missing values, and it contributes 201 missing records.

The heatmap shows a very narrow missingness pattern rather than scattered nulls.

Duplicate rows are absent, which simplifies downstream preprocessing.

The categorical variables have a few rare levels such as Other and Never_worked.

Conclusion

The data quality is good overall, with BMI missingness as the main issue.

Question 4

Aim

Analyze the target variable.

Program

```
target_df = pd.DataFrame({
    'count': target_distribution(df),
    'percentage': target_percentages(df),
})
display(target_df)
create_target_plots(df)
```

Output

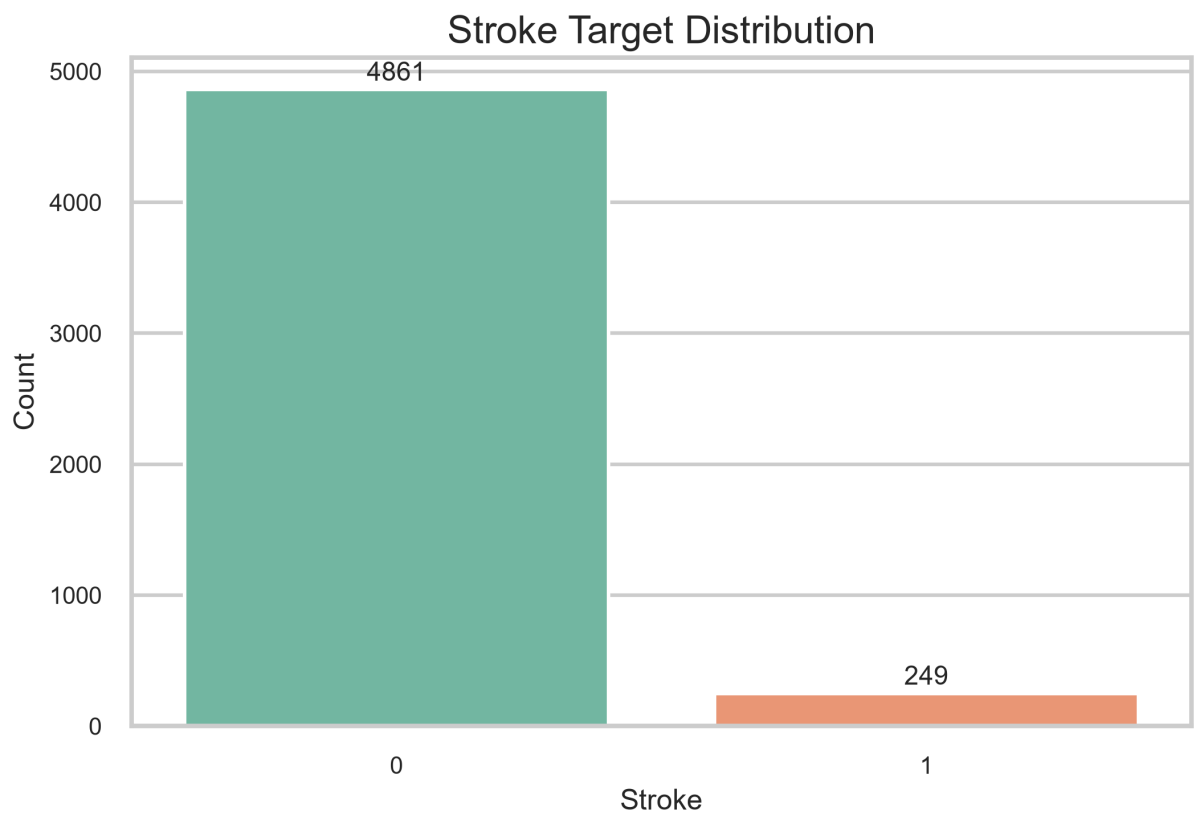


Figure 17. Stroke count plot.

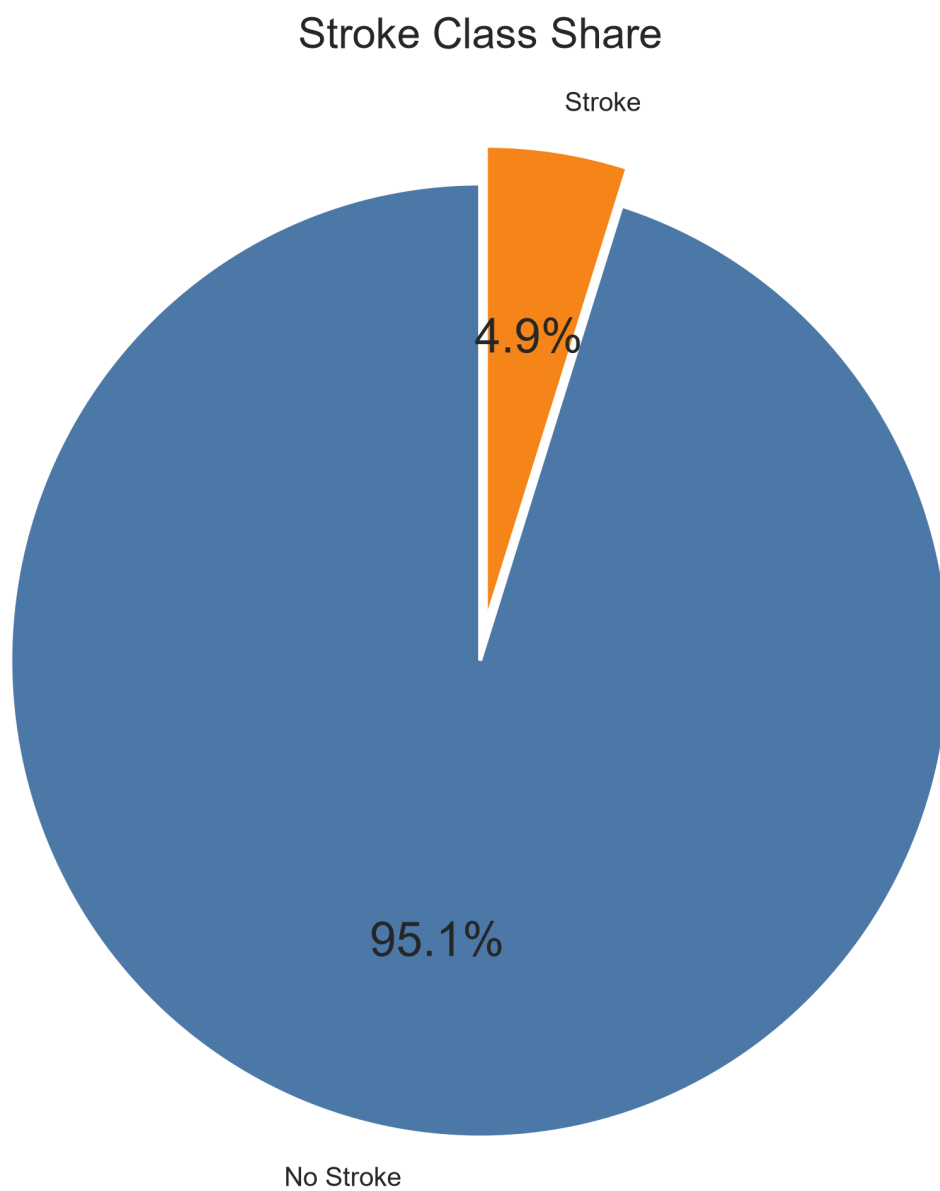


Figure 18. Stroke pie chart.

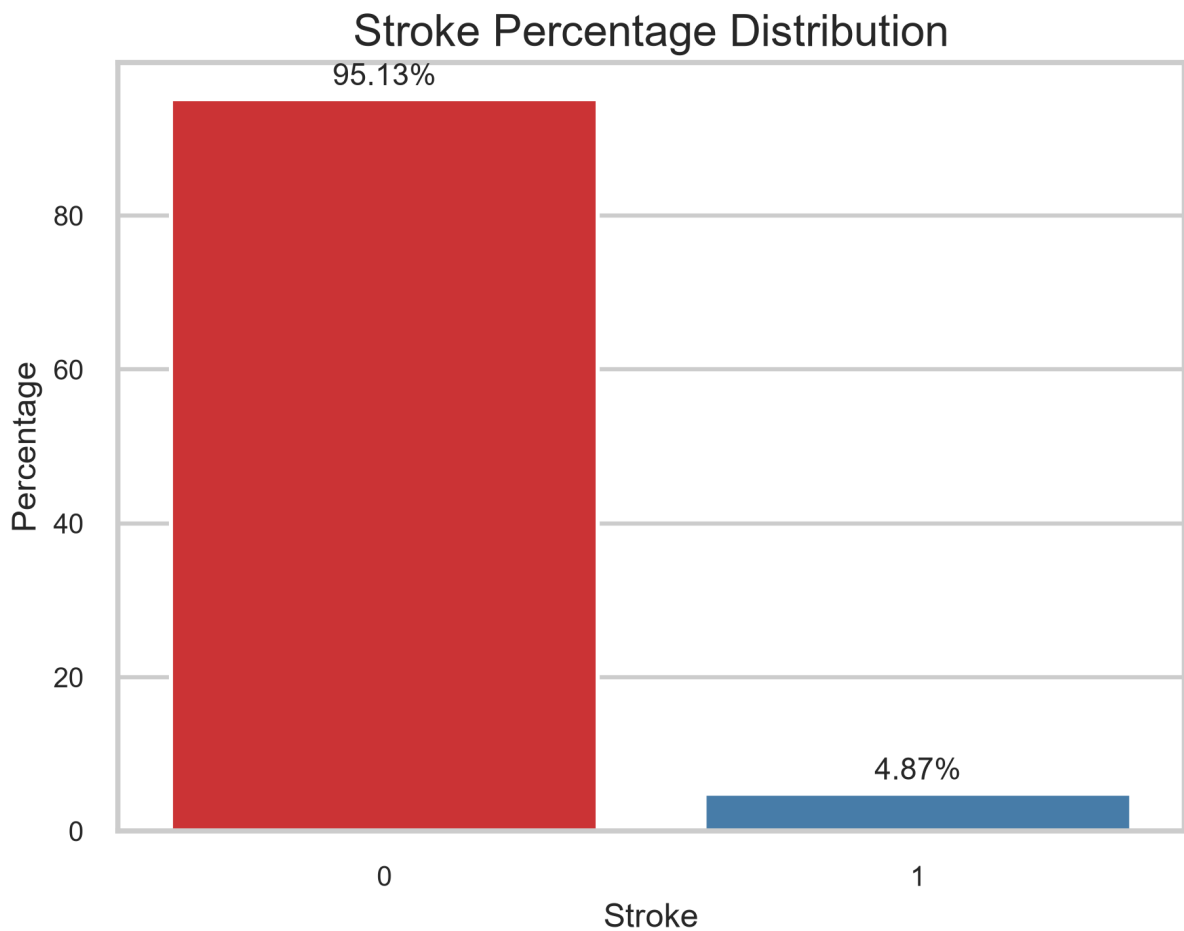


Table 1. Stroke class percentage distribution.

Stroke	Count	Percentage
0	4861	95.13%
1	249	4.87%

Analysis

Stroke class 0 contains 4861 records (95.13%).

Stroke class 1 contains 249 records (4.87%).

The class distribution is highly imbalanced and accuracy alone would be misleading.

The pie chart and count plot both show that stroke is a minority class.

Conclusion

Stroke is a severely imbalanced target variable.

Question 5

Aim

Analyze numerical features.

Program

```
display(statistical_summary(df))  
create_numeric_distribution_plots(df)
```

Output

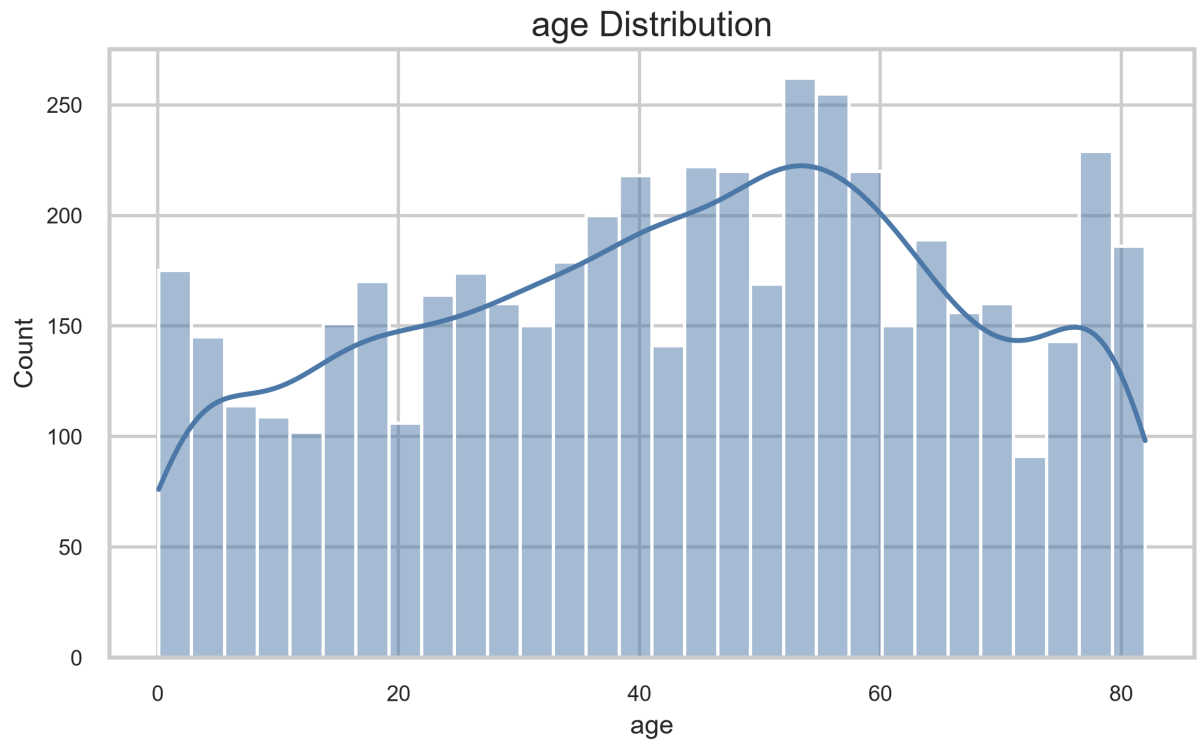


Figure 20. Age histogram.

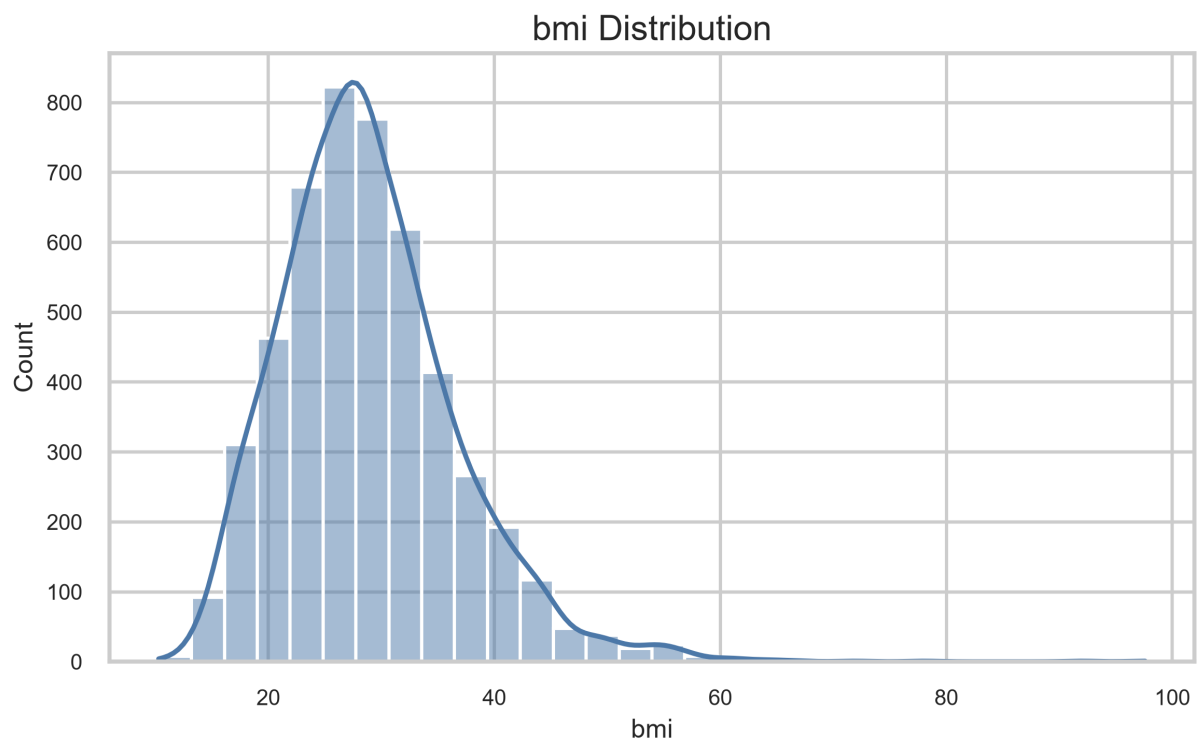


Figure 21. BMI histogram.

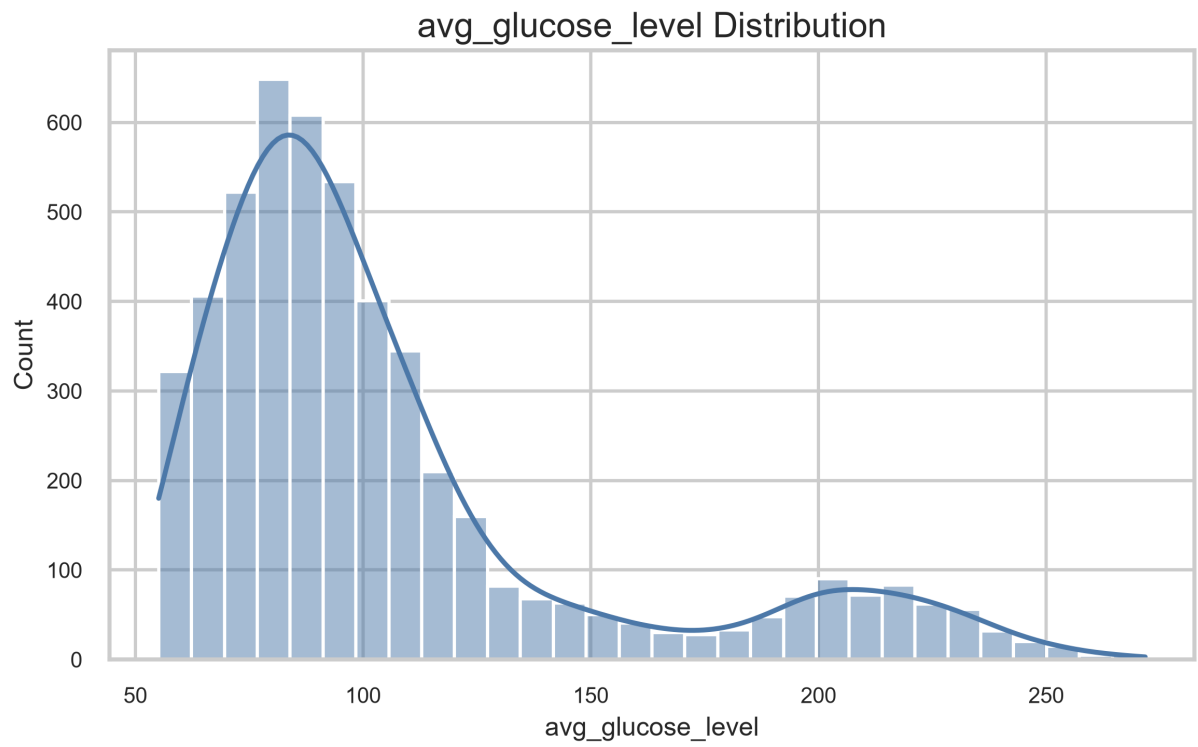


Figure 22. Glucose histogram.

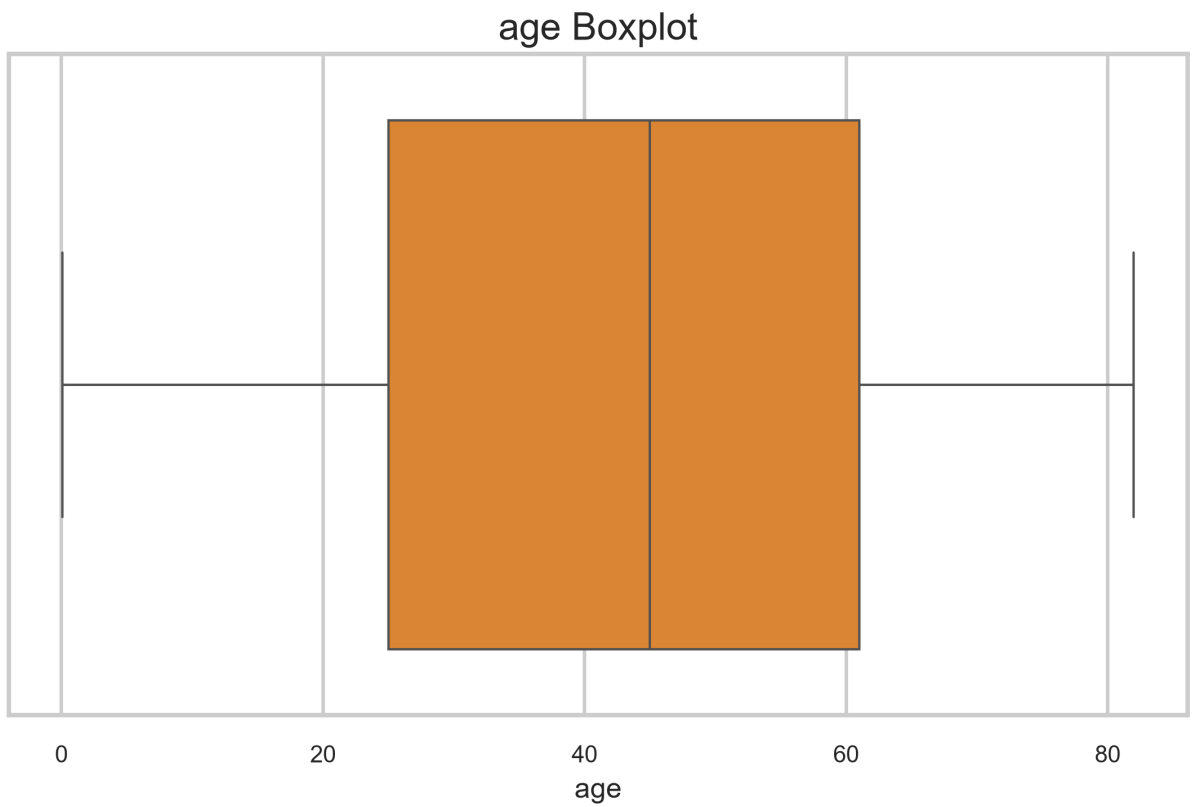


Figure 23. Age boxplot.

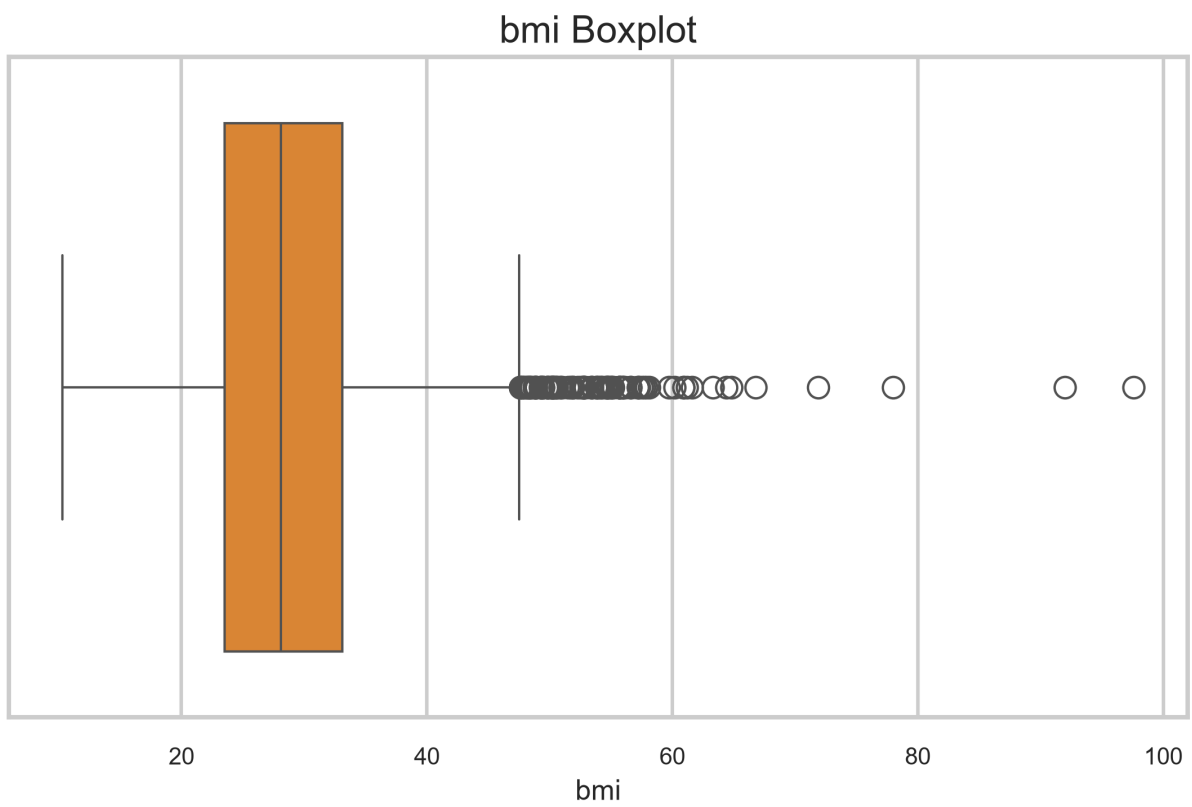


Figure 24. BMI boxplot.

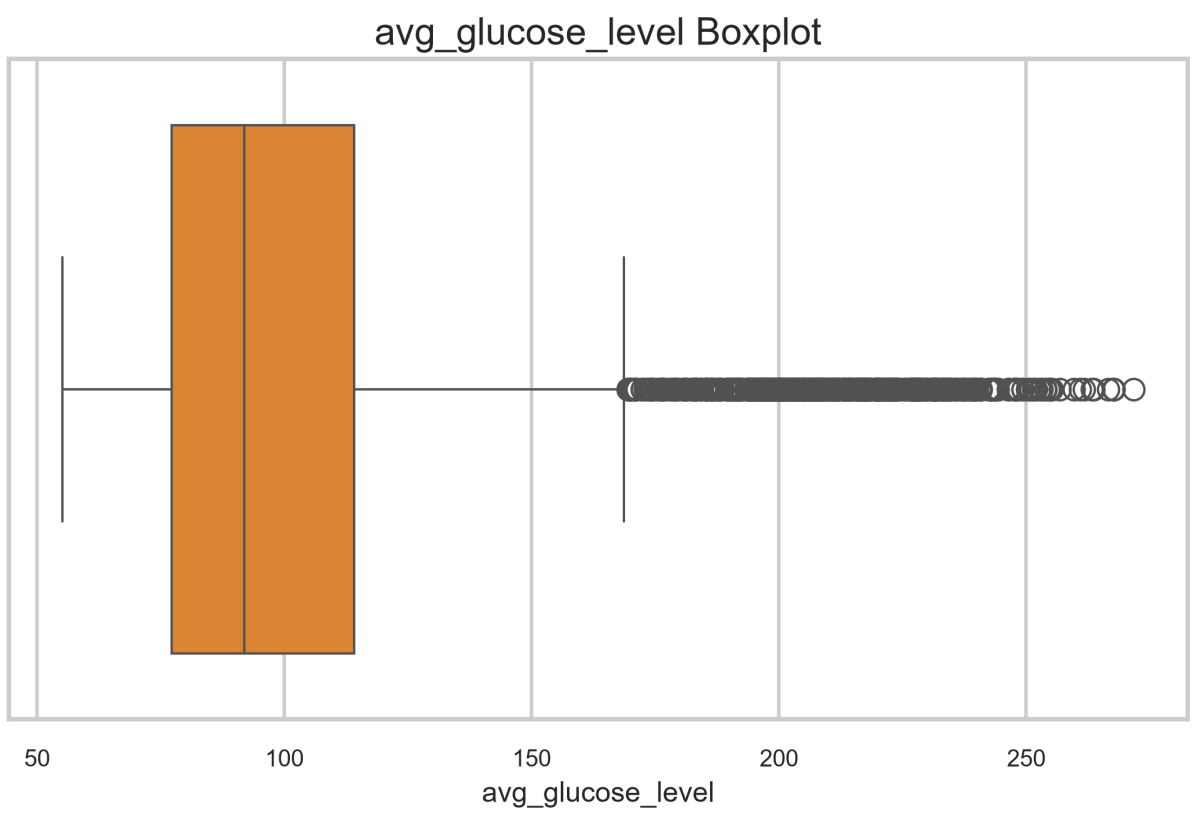


Figure 25. Glucose boxplot.

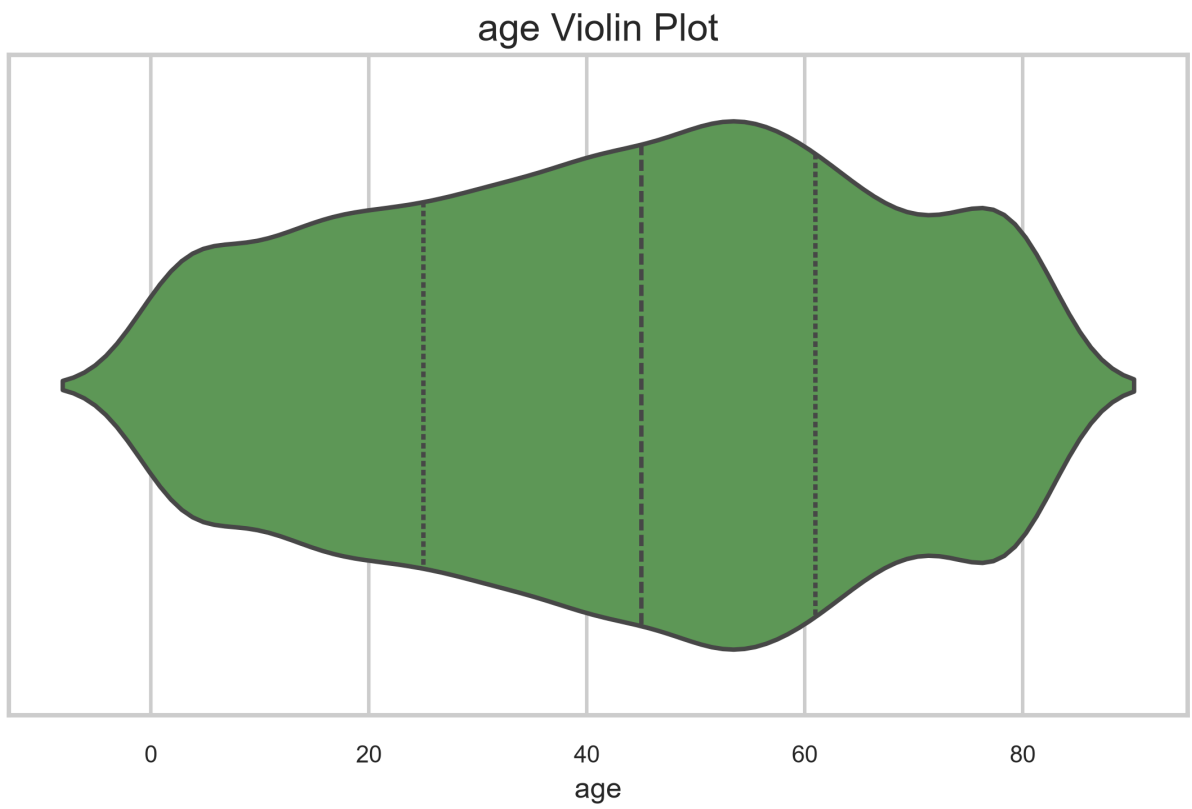


Figure 26. Age violin plot.

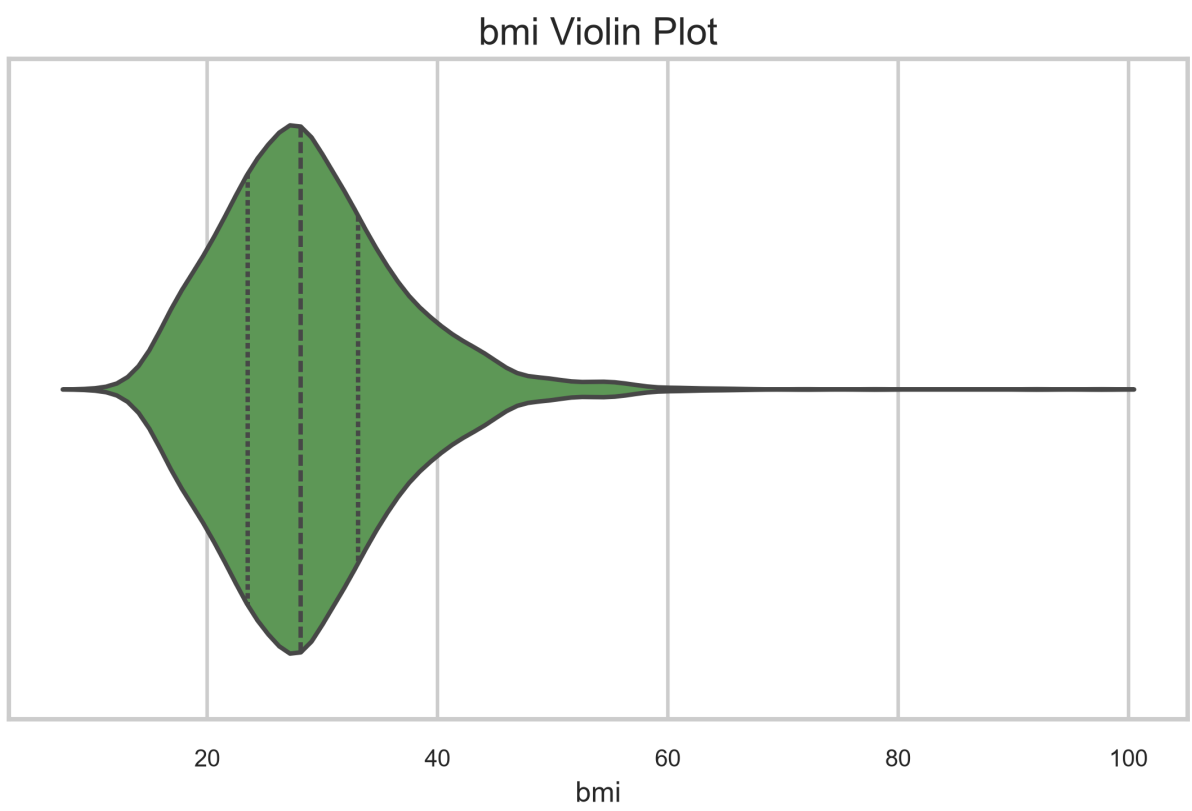


Figure 27. BMI violin plot.

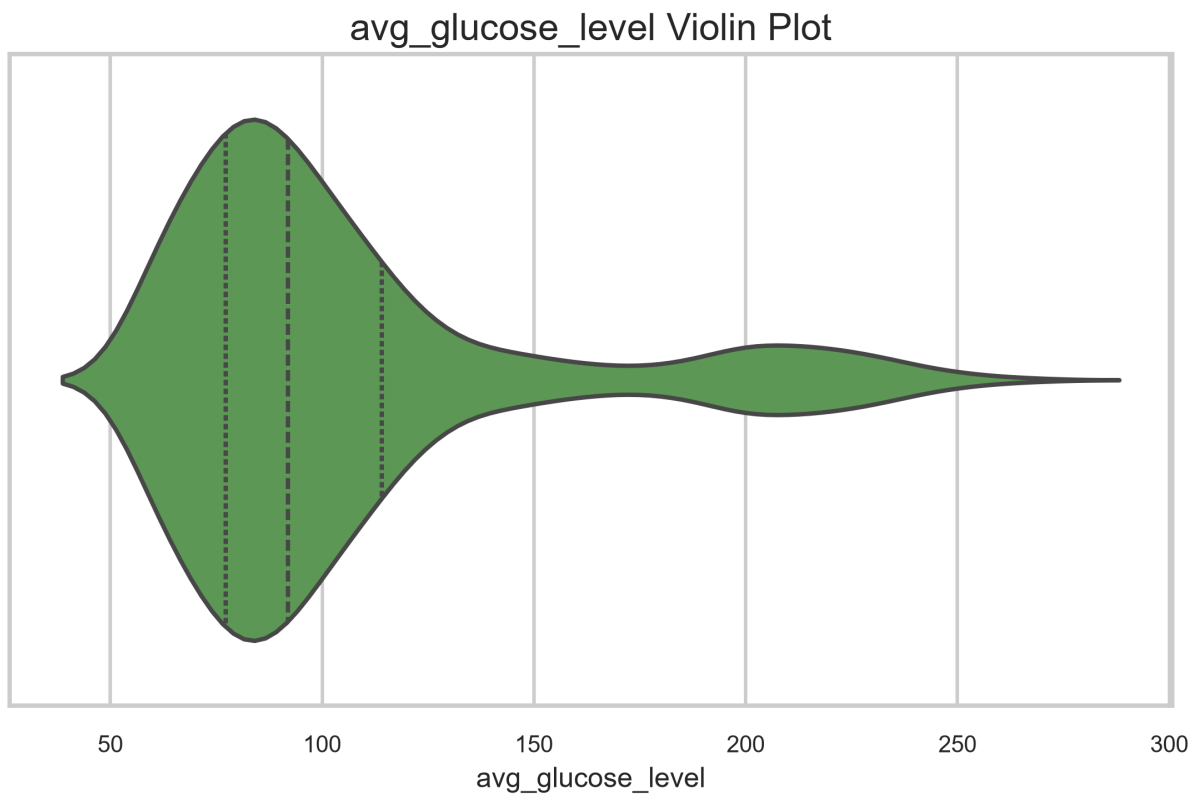


Figure 28. Glucose violin plot.

Analysis

Age is broadly spread and visually close to a symmetric distribution.

BMI is right-skewed with a long upper tail.

Average glucose level has the strongest skewness and the most extreme values.

The violin plots confirm that the central mass of BMI and glucose is concentrated below the upper tails.

Conclusion

The numerical variables are skewed and require preprocessing.

Question 6

Aim

Analyze categorical variables.

Program

```
create_categorical_countplots(df)
for column in ['gender', 'work_type', 'ever_married', 'Residence_type', 'smoking_status', 'hypertens
    print(f'\n{column}:')
    display(df[column].value_counts(dropna=False).to_frame(name='count'))
```

Output

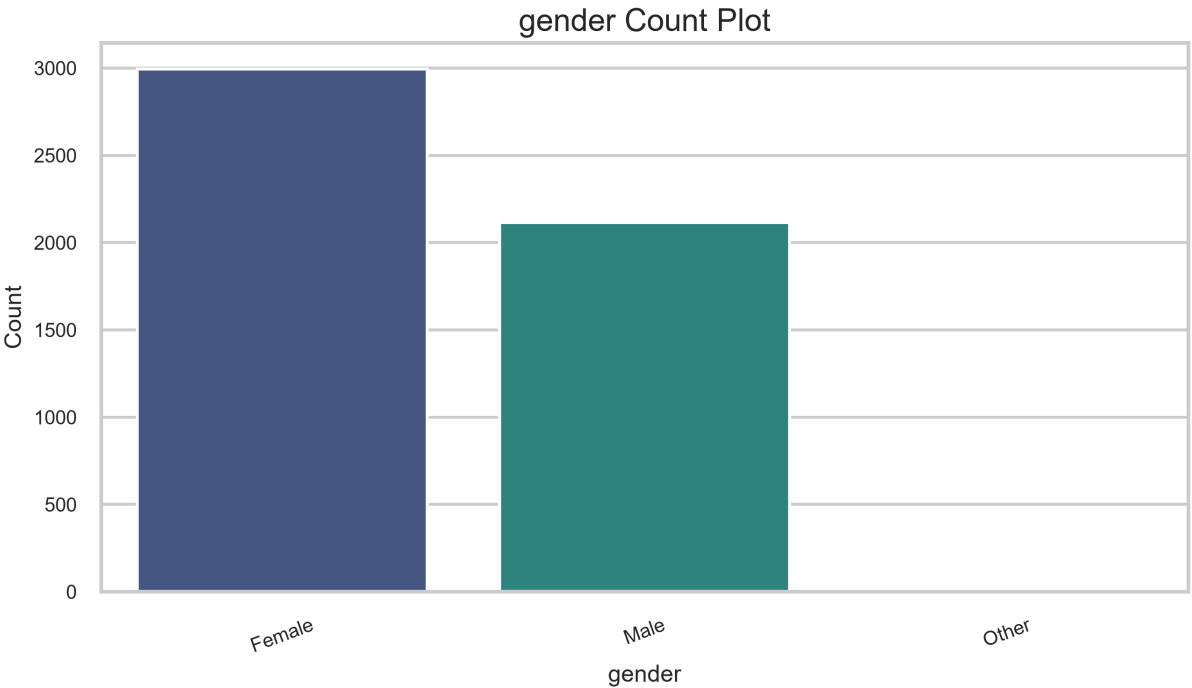


Figure 29. Gender count plot.

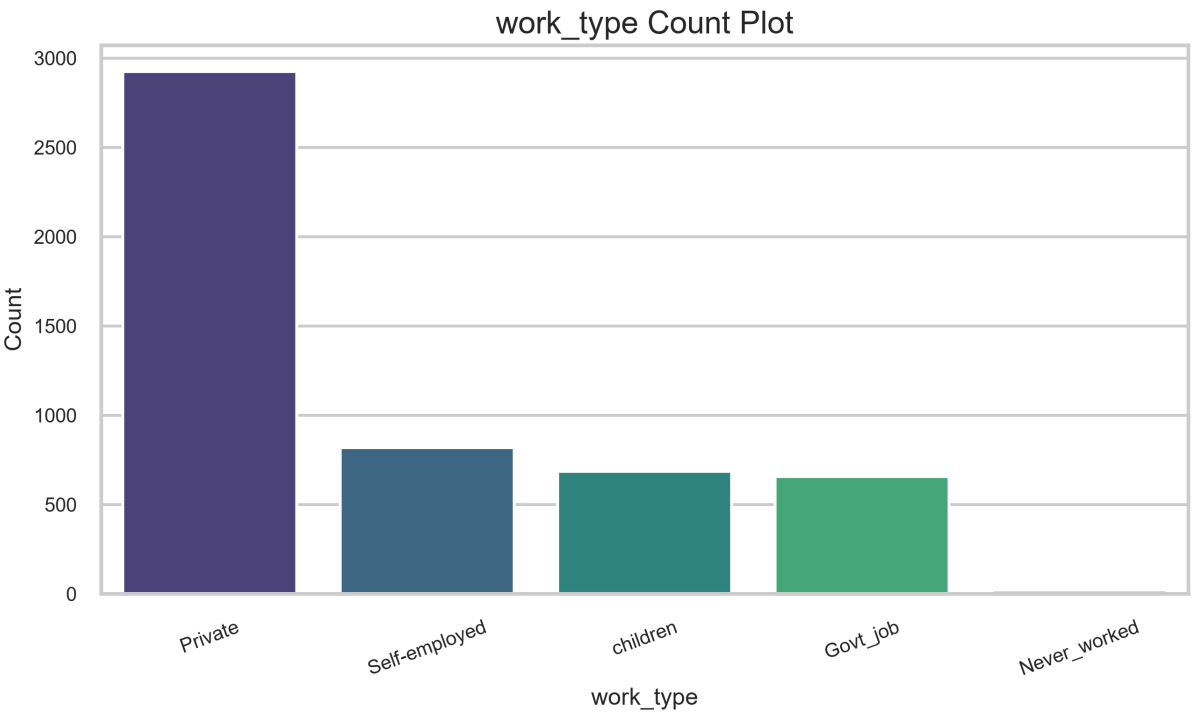


Figure 30. Work type count plot.

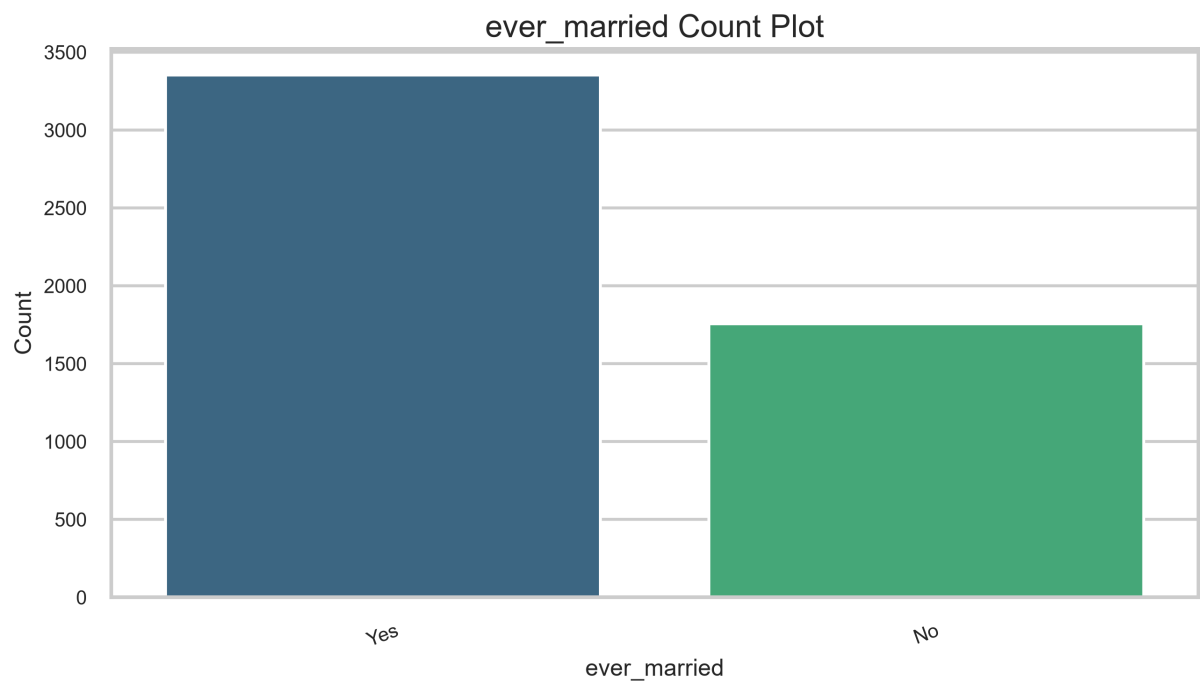


Figure 31. Ever married count plot.

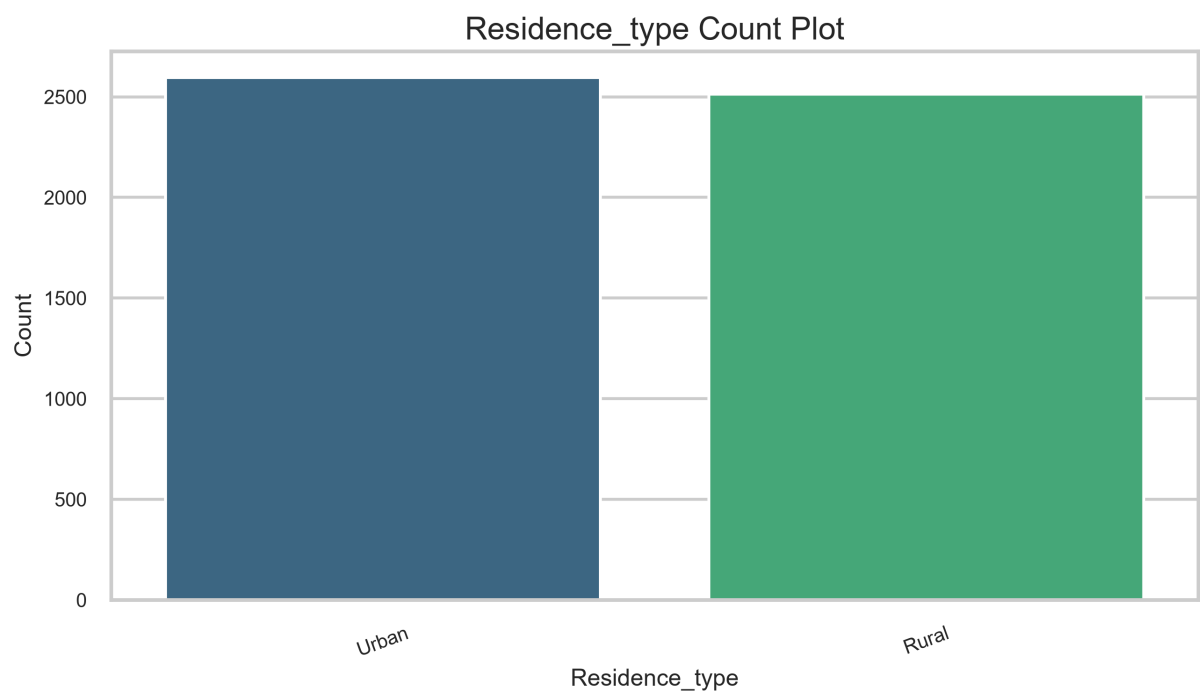


Figure 32. Residence type count plot.

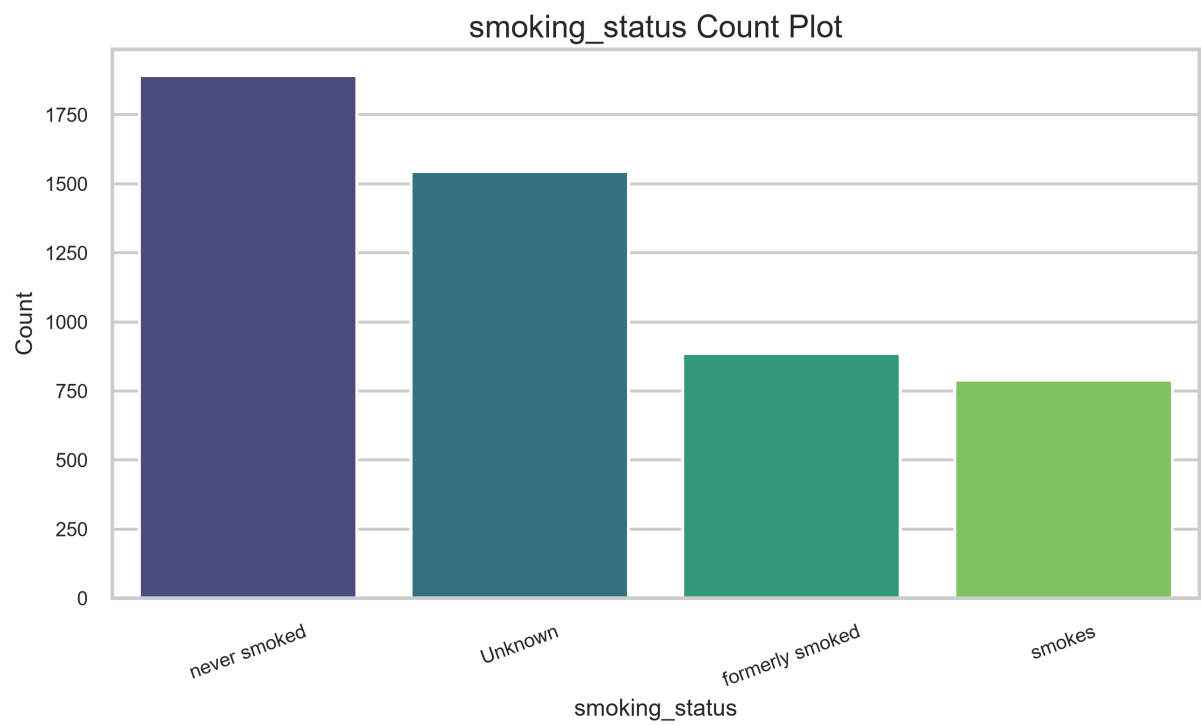


Figure 33. Smoking status count plot.

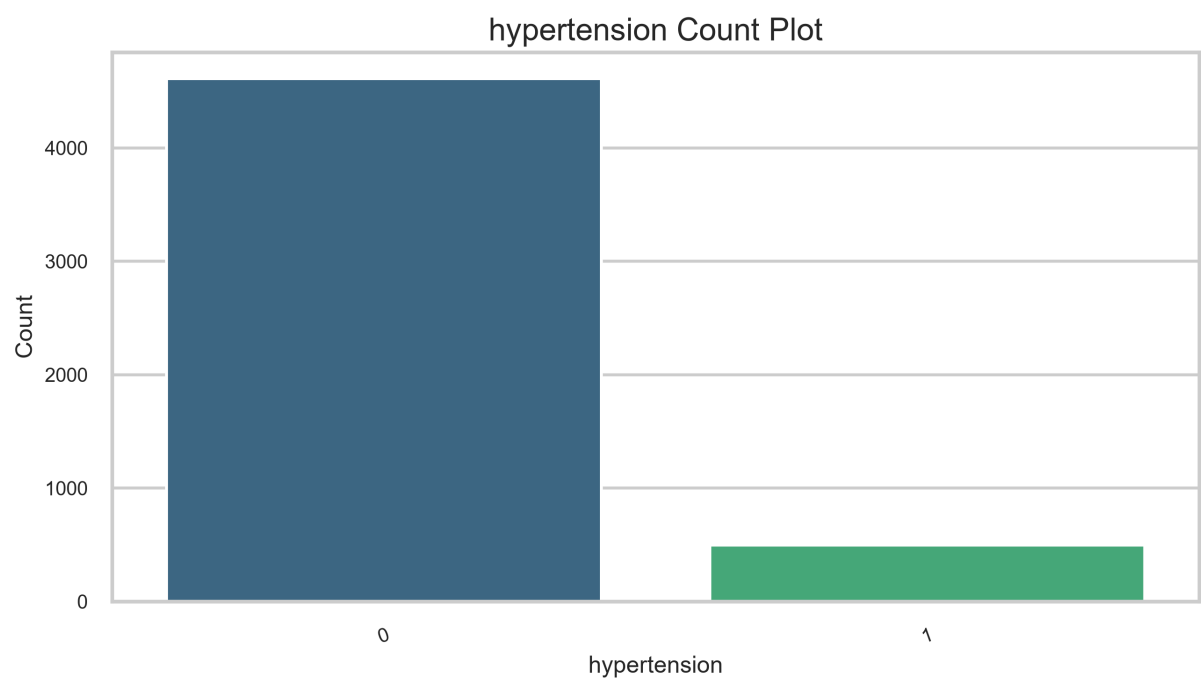


Figure 34. Hypertension count plot.

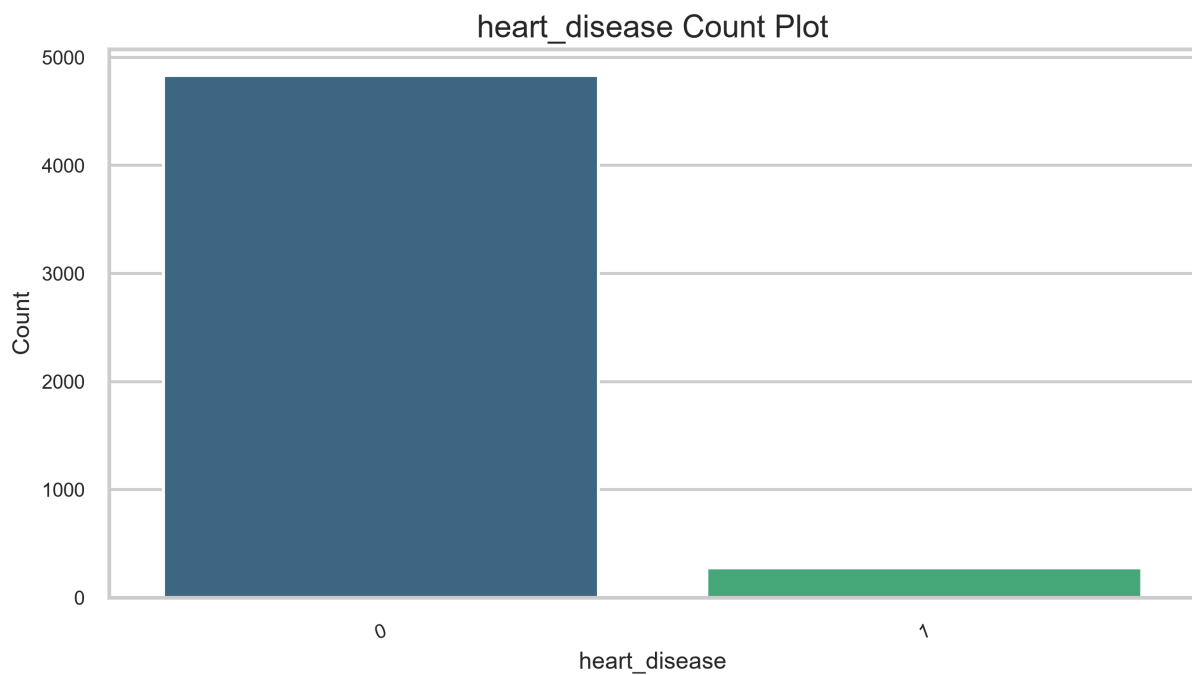


Figure 35. Heart disease count plot.

Analysis

Female records are more common than male records, with a single Other record.

Private work type is the largest category, followed by Self-employed and children.

Residence type is nearly balanced between Urban and Rural.

Smoking status includes a large Unknown group, which should be handled carefully later.

Conclusion

The categorical variables are clean but contain a few rare levels.

Question 7

Aim

Perform bivariate analysis.

Program

```
create_bivariate_plots(df)
```

Output

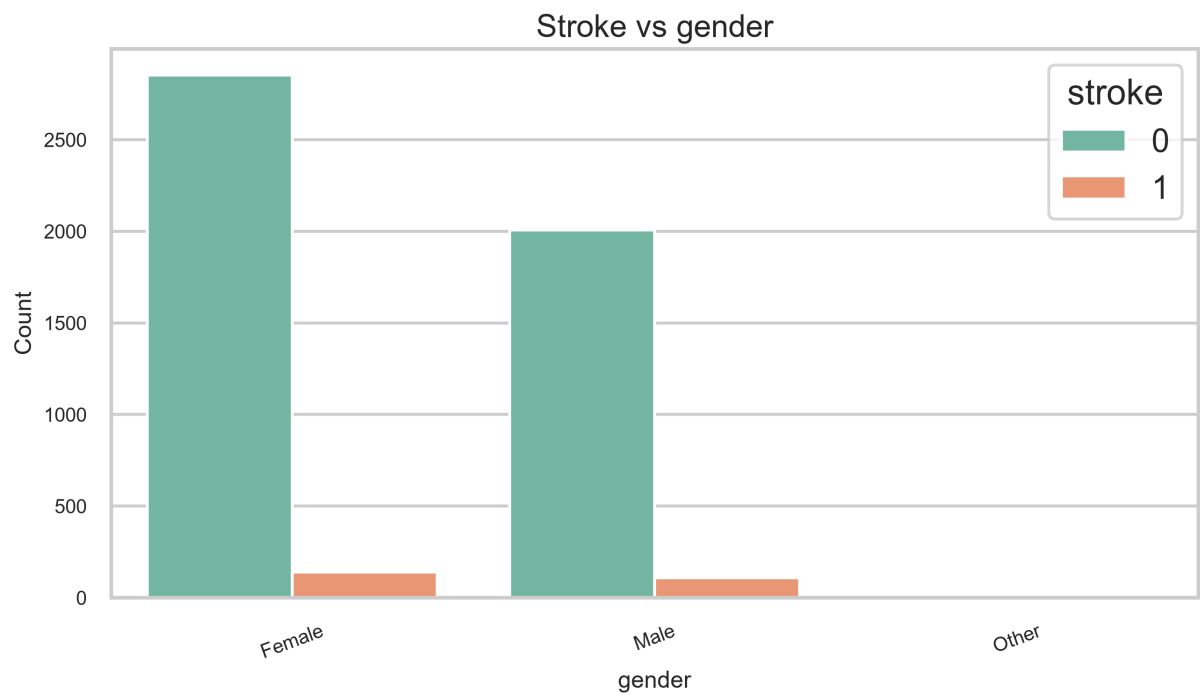


Figure 36. Stroke by gender.

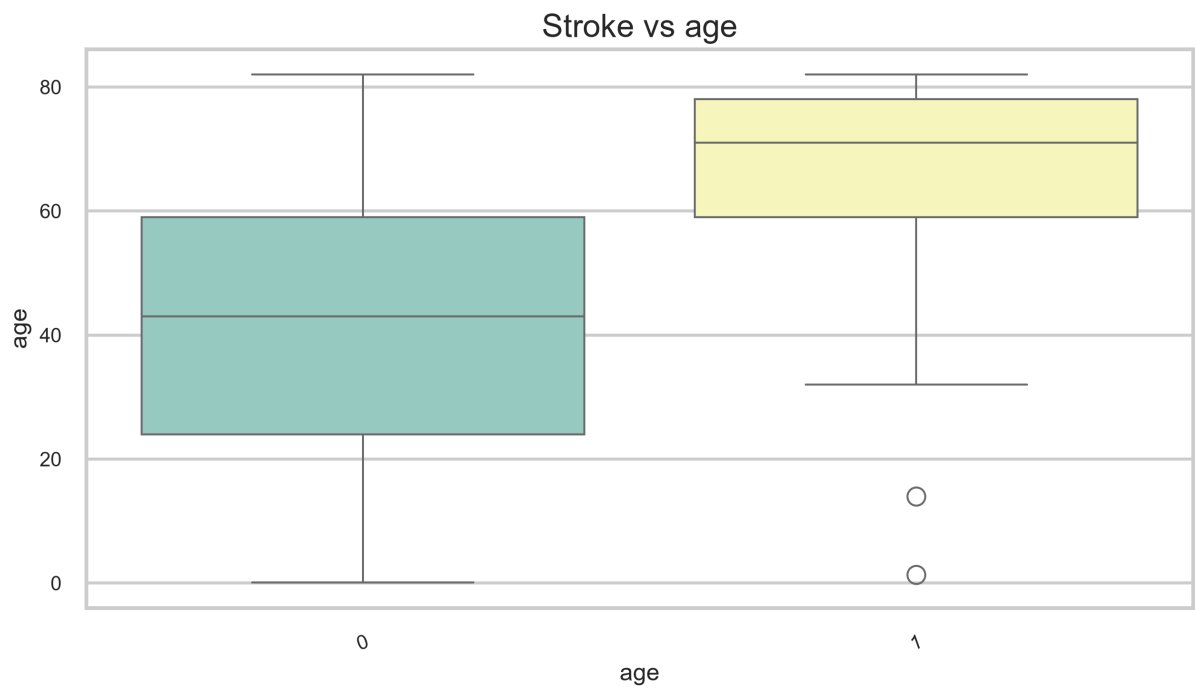


Figure 37. Stroke by age.

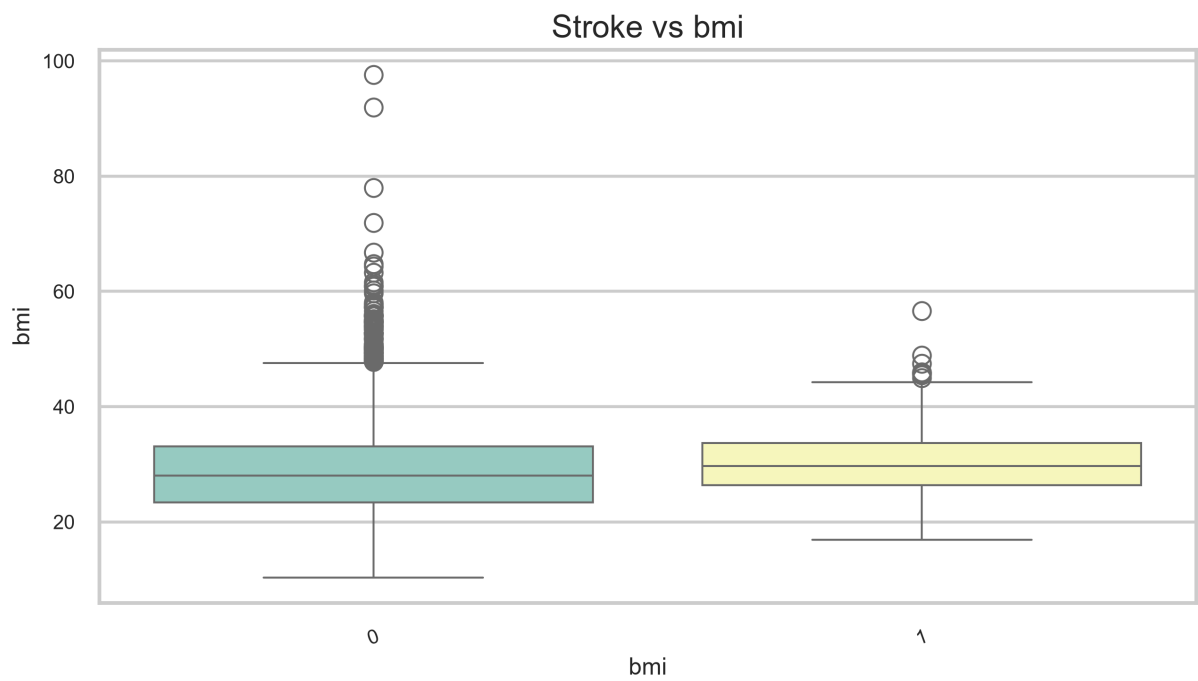


Figure 38. Stroke by BMI.

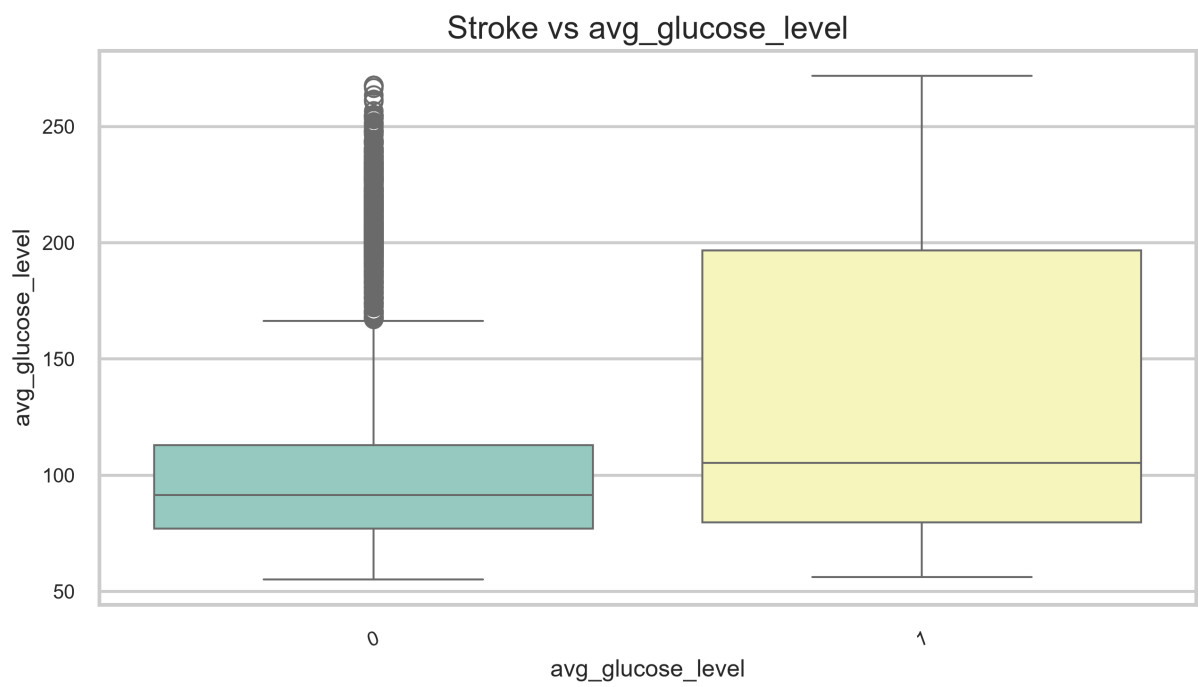


Figure 39. Stroke by glucose level.

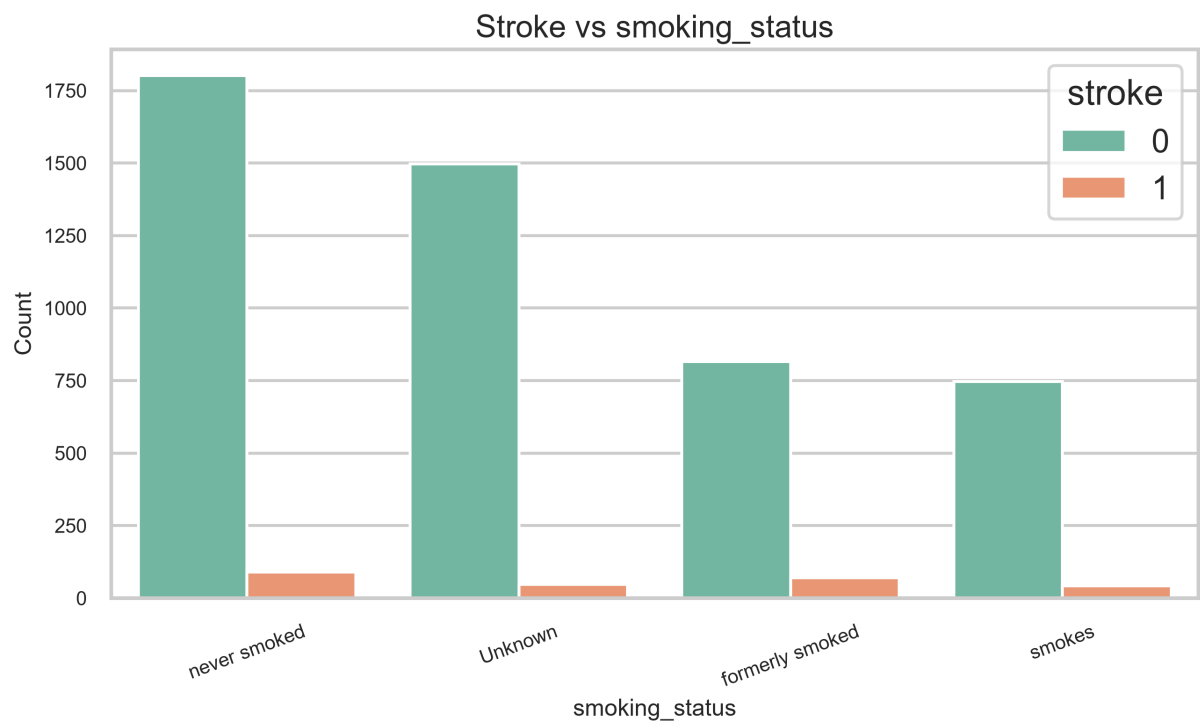


Figure 40. Stroke by smoking status.

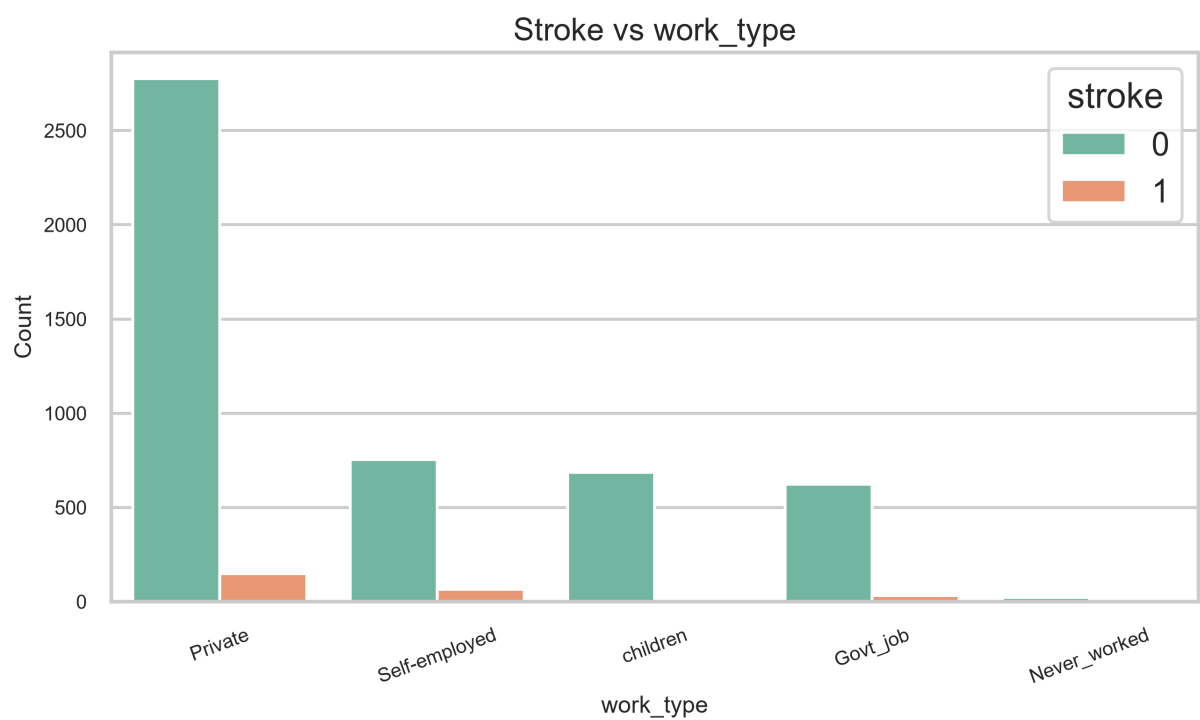


Figure 41. Stroke by work type.

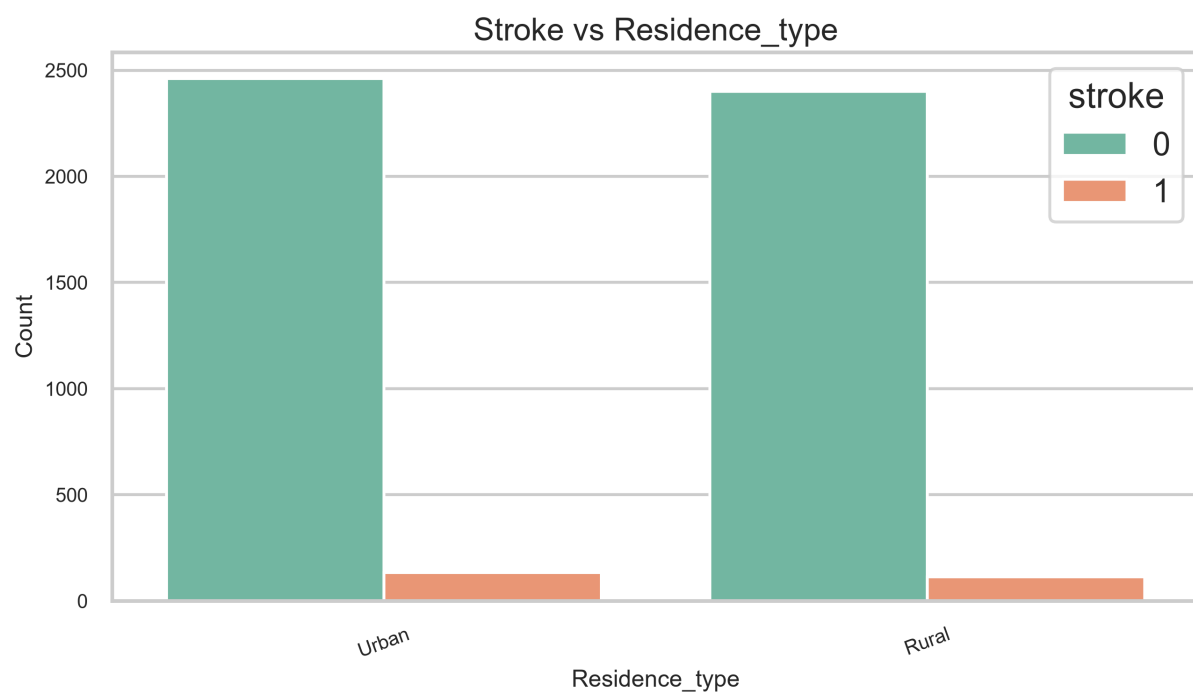


Figure 42. Stroke by residence type.

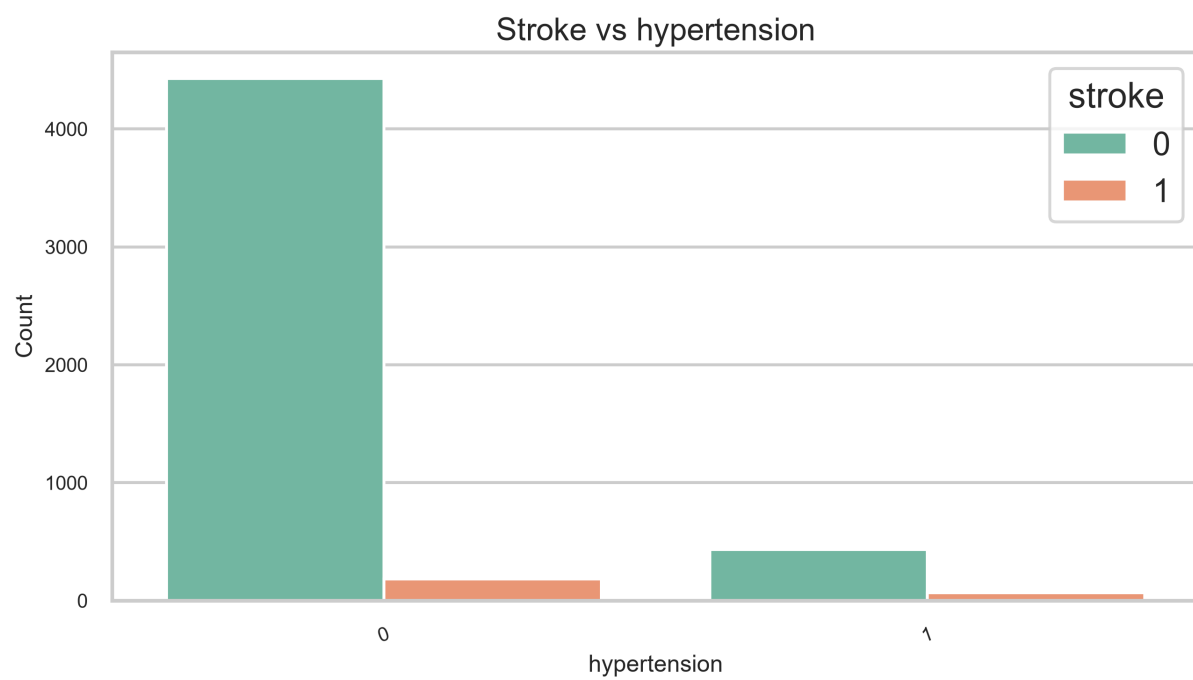


Figure 43. Stroke by hypertension.

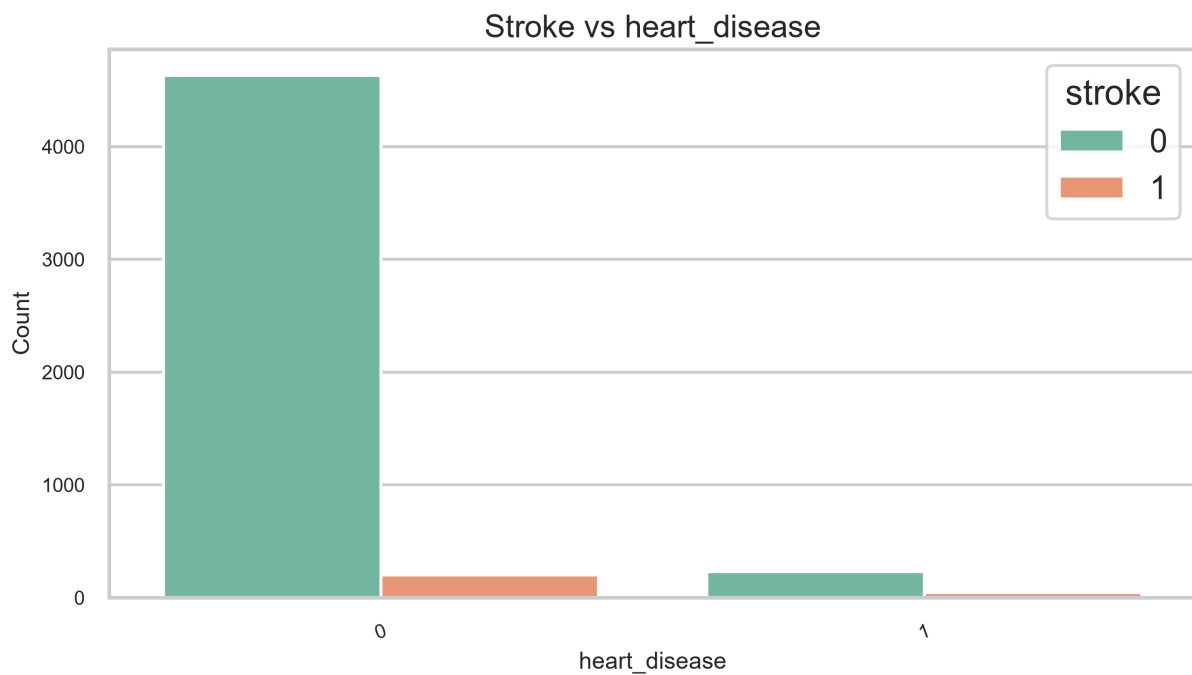


Figure 44. Stroke by heart disease.

Analysis

Stroke is visibly more common among older patients.

Patients with hypertension or heart disease show a higher stroke proportion.

Former smokers and self-employed patients show elevated stroke proportions.

Residence type has a much weaker relationship with stroke than the clinical factors.

Conclusion

Older age, hypertension, heart disease, and glucose are the clearest bivariate signals.

Question 8

Aim

Perform correlation analysis.

Program

```
corr = create_correlation_heatmap(df)
display(corr['stroke'].sort_values(ascending=False).to_frame(name='correlation_to_stroke'))
```

Output

Correlation Matrix

	id	age	hypertension	heart_disease	glucose_level	bmi	stroke	gender_Female	gender_Male	gender_Other	ever_married	work_type_Govt_job	work_type_Never_worked	work_type_Private	work_type_Self-employed	work_type_children	Residence_type_Rural	Residence_type_Urban	smoking_status_Unknown	smoking_status_formerly smoked	smoking_status_never smoked	smoking_status_smokes	
id	1.0	0.0	0.0	-0.0	0.0	0.0	0.01	-0.0	0.0	0.01	-0.01	0.01	-0.0	0.01	-0.02	-0.01	0.0	-0.0	-0.02	0.01	0.01	0.0	
age	0.0	1.0	0.28	0.26	0.24	0.33	0.25	0.03	-0.03	-0.01	-0.68	0.68	0.13	-0.06	0.12	0.33	-0.63	-0.01	0.01	-0.38	0.24	0.12	0.07
hypertension	0.0	0.28	1.0	0.11	0.17	0.17	0.13	-0.02	0.02	-0.0	-0.16	0.16	0.02	-0.02	-0.01	0.12	-0.13	0.01	-0.01	-0.14	0.06	0.07	0.03
heart_disease	-0.0	0.26	0.11	1.0	0.16	0.04	0.13	-0.09	0.09	-0.0	-0.11	0.11	0.0	-0.02	0.0	0.09	-0.09	-0.0	0.0	-0.07	0.07	-0.02	0.04
avg_glucose_level	0.0	0.24	0.17	0.16	1.0	0.18	0.13	-0.05	0.05	0.01	-0.16	0.16	0.01	-0.01	0.02	0.06	-0.1	0.0	-0.0	-0.1	0.07	0.02	0.02
bmi	0.0	0.33	0.17	0.04	0.18	1.0	0.04	0.03	-0.03	-0.01	-0.34	0.34	0.08	-0.03	0.21	0.07	-0.45	0.0	-0.0	-0.27	0.11	0.11	0.09
stroke	0.01	0.25	0.13	0.13	0.13	0.04	1.0	-0.01	0.01	-0.0	-0.11	0.11	0.0	-0.01	0.01	0.06	-0.08	-0.02	0.02	-0.06	0.06	-0.0	0.01
gender_Female	-0.0	0.03	-0.02	-0.09	-0.05	0.03	-0.01	1.0	-1.0	-0.02	-0.03	0.03	0.02	-0.01	0.03	0.03	-0.09	-0.01	0.01	-0.06	-0.04	0.1	-0.01
gender_Male	0.0	-0.03	0.02	0.09	0.05	-0.03	0.01	-1.0	1.0	-0.01	0.03	-0.03	-0.02	0.01	-0.03	-0.03	0.09	0.01	-0.01	0.06	0.04	-0.1	0.01
gender_Other	0.01	-0.01	-0.0	-0.0	0.01	-0.01	-0.0	-0.02	-0.01	1.0	0.02	-0.02	-0.01	-0.0	0.01	-0.01	-0.01	0.01	-0.01	-0.01	0.03	-0.01	-0.01
ever_married_No	-0.01	-0.68	-0.16	-0.11	-0.16	-0.34	-0.11	-0.03	0.03	0.02	1.0	-1.0	-0.13	0.09	-0.15	-0.19	0.54	0.01	-0.01	0.33	-0.17	-0.1	-0.11
ever_married_Yes	0.01	0.68	0.16	0.11	0.16	0.34	0.11	0.03	-0.03	-0.02	-1.0	1.0	0.13	-0.09	0.15	0.19	-0.54	-0.01	0.01	-0.33	0.17	0.1	0.11
work_type_Govt_job	-0.0	0.13	0.02	0.0	0.01	0.08	0.0	0.02	-0.02	-0.01	-0.13	0.13	1.0	-0.03	-0.44	-0.17	-0.15	-0.01	0.01	-0.1	0.03	0.05	0.03
work_type_Never_worked	0.01	-0.08	-0.02	-0.02	-0.01	-0.03	-0.01	-0.01	0.01	-0.0	0.09	-0.09	-0.03	1.0	-0.08	-0.03	-0.03	-0.02	0.02	0.01	-0.03	0.04	-0.03
work_type_Private	0.02	0.12	-0.01	0.0	0.02	0.21	0.01	0.03	-0.03	0.01	-0.15	0.15	-0.44	-0.08	1.0	-0.51	-0.46	0.02	-0.02	-0.21	0.03	0.1	0.1
work_type_Self-employed	-0.02	0.33	0.12	0.09	0.06	0.07	0.06	0.03	-0.03	-0.01	-0.19	0.19	-0.17	-0.03	-0.51	1.0	-0.17	-0.01	0.01	-0.11	0.09	0.03	-0.0
work_type_children	-0.01	-0.63	-0.13	-0.09	-0.1	-0.45	-0.08	-0.09	0.09	-0.01	0.54	-0.54	-0.15	-0.03	-0.46	-0.17	1.0	0.0	-0.0	0.51	-0.16	-0.24	-0.17
Residence_type_Rural	0.0	-0.01	0.01	-0.0	0.0	0.0	-0.02	-0.01	0.01	0.01	0.01	-0.01	-0.01	-0.02	0.02	-0.01	0.0	1.0	-1.0	0.0	-0.01	0.02	-0.03
Residence_type_Urban	-0.0	0.01	-0.01	0.0	-0.0	-0.0	0.02	0.01	-0.01	-0.01	-0.01	0.01	0.01	0.02	-0.02	0.01	-0.0	-1.0	1.0	-0.0	0.01	-0.02	0.03
smoking_status_Unknown	-0.02	-0.38	-0.14	-0.07	-0.1	-0.27	-0.06	-0.06	0.06	-0.01	0.33	-0.33	-0.1	0.01	-0.21	-0.11	0.51	0.0	-0.0	1.0	-0.3	-0.5	-0.28
smoking_status_formerly smoked	0.01	0.24	0.06	0.07	0.07	0.11	0.06	-0.04	0.04	0.03	-0.17	0.17	0.03	-0.03	0.03	0.09	-0.16	-0.01	0.01	-0.3	1.0	-0.35	-0.2
smoking_status_never smoked	0.01	0.12	0.07	-0.02	0.02	0.11	-0.0	0.1	-0.1	-0.01	-0.1	0.1	0.05	0.04	0.1	0.03	-0.24	0.02	-0.02	-0.5	-0.35	1.0	-0.33
smoking_status_smokes	0.0	0.07	0.03	0.04	0.02	0.09	0.01	-0.01	0.01	-0.01	-0.11	0.11	0.03	-0.03	0.1	-0.0	-0.17	-0.03	0.03	-0.28	-0.2	-0.33	1.0

Figure 45. Correlation matrix.

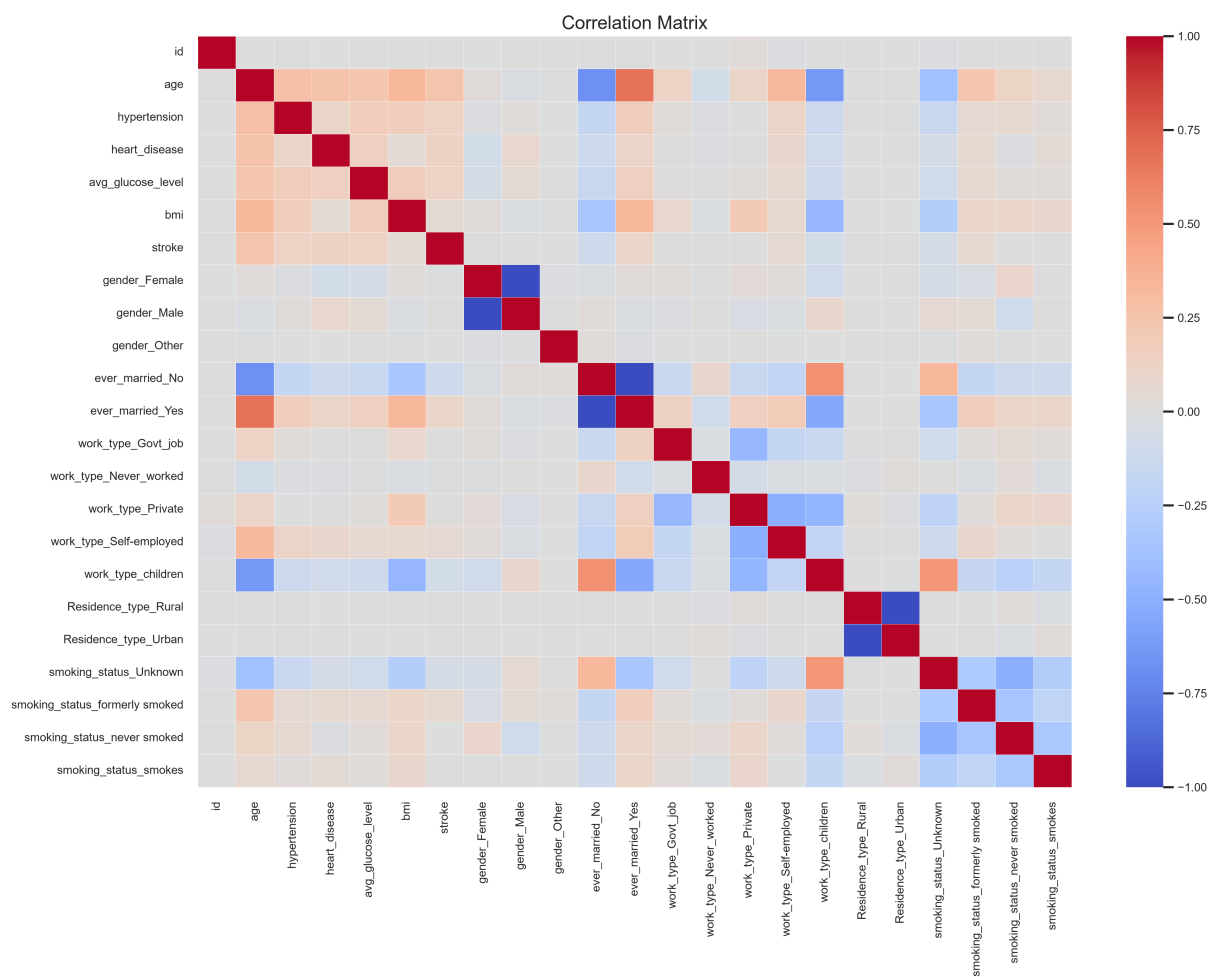


Figure 46. Correlation heatmap.

Feature	Correlation with Stroke
stroke	1.000
age	0.245
heart_disease	0.135
avg_glucose_level	0.132
hypertension	0.128
ever_married_Yes	0.108
smoking_status_formerly smoked	0.065
work_type_Self-employed	0.062
bmi	0.042
Residence_type_Urban	0.015
work_type_Private	0.012
gender_Male	0.009
smoking_status_smokes	0.009
id	0.006
work_type_Govt_job	0.003
gender_Other	-0.003

Feature	Correlation with Stroke
smoking_status_never smoked	-0.004
gender_Female	-0.009
work_type_Never_worked	-0.015
Residence_type_Rural	-0.015
smoking_status_Unknown	-0.056
work_type_children	-0.084
ever_married_No	-0.108

Analysis

Age has the strongest positive correlation with stroke among the encoded variables.

Heart disease, average glucose level, hypertension, and marital status also show positive association.

Being unmarried and being in the children work type are negatively correlated with stroke.

The heatmap shows moderate, not extreme, relationships overall.

Conclusion

Age is the strongest positive correlation with stroke.

Question 9

Aim

Perform outlier detection and distribution analysis.

Program

```
display(outlier_summary(df))
create_outlier_plots(df)

display(normality_summary(df))
create_distribution_analysis_plots(df)
```

Output

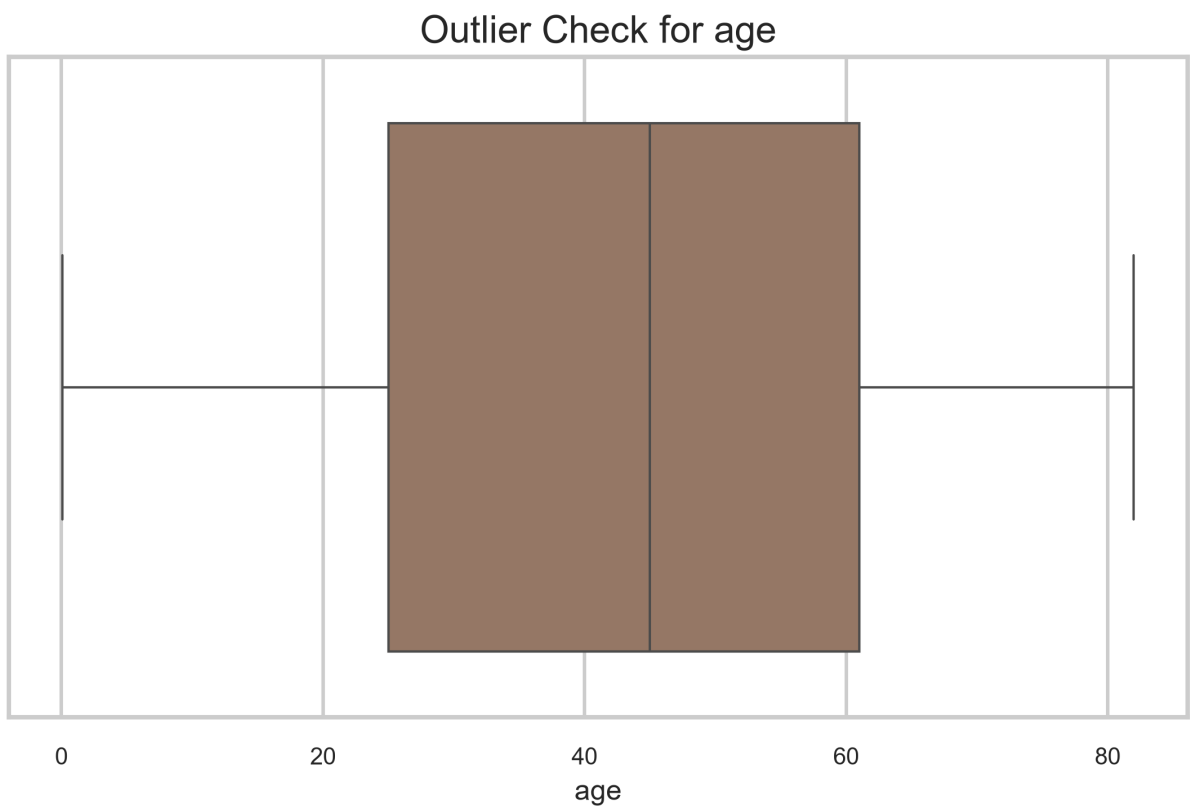


Figure 47. Age outlier boxplot.

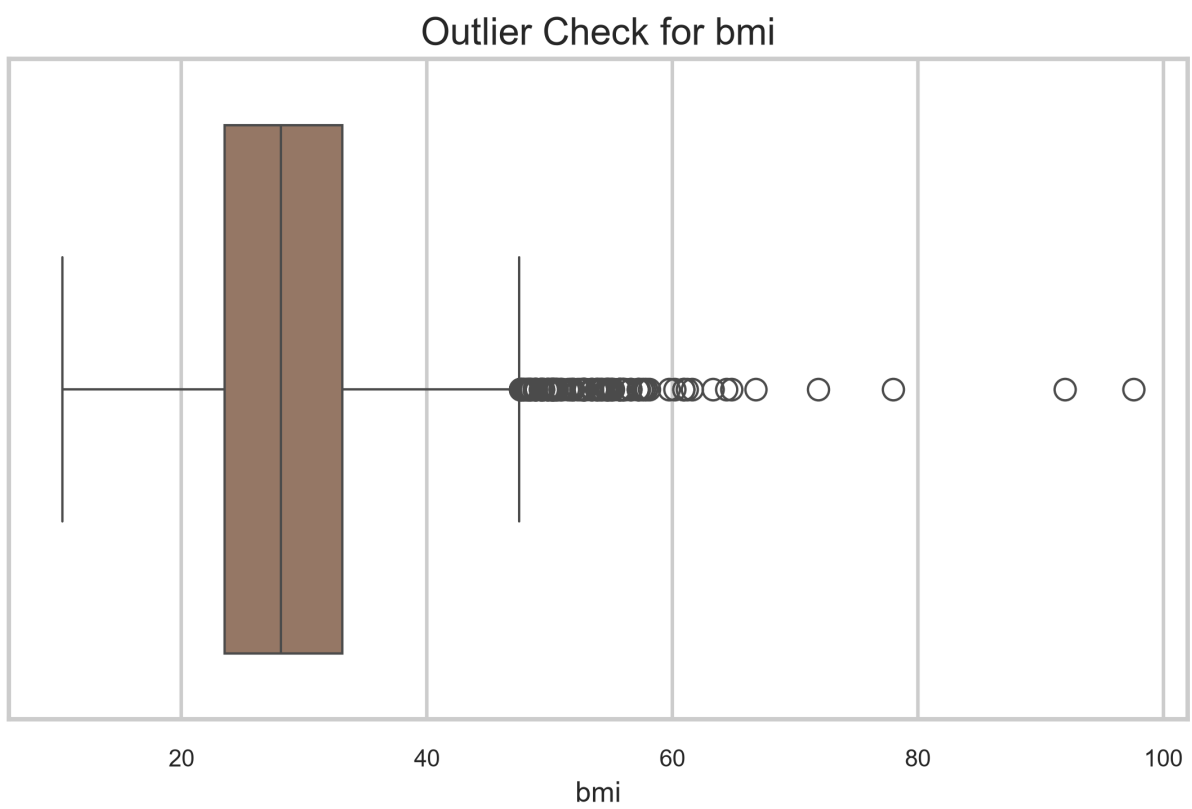


Figure 48. BMI outlier boxplot.

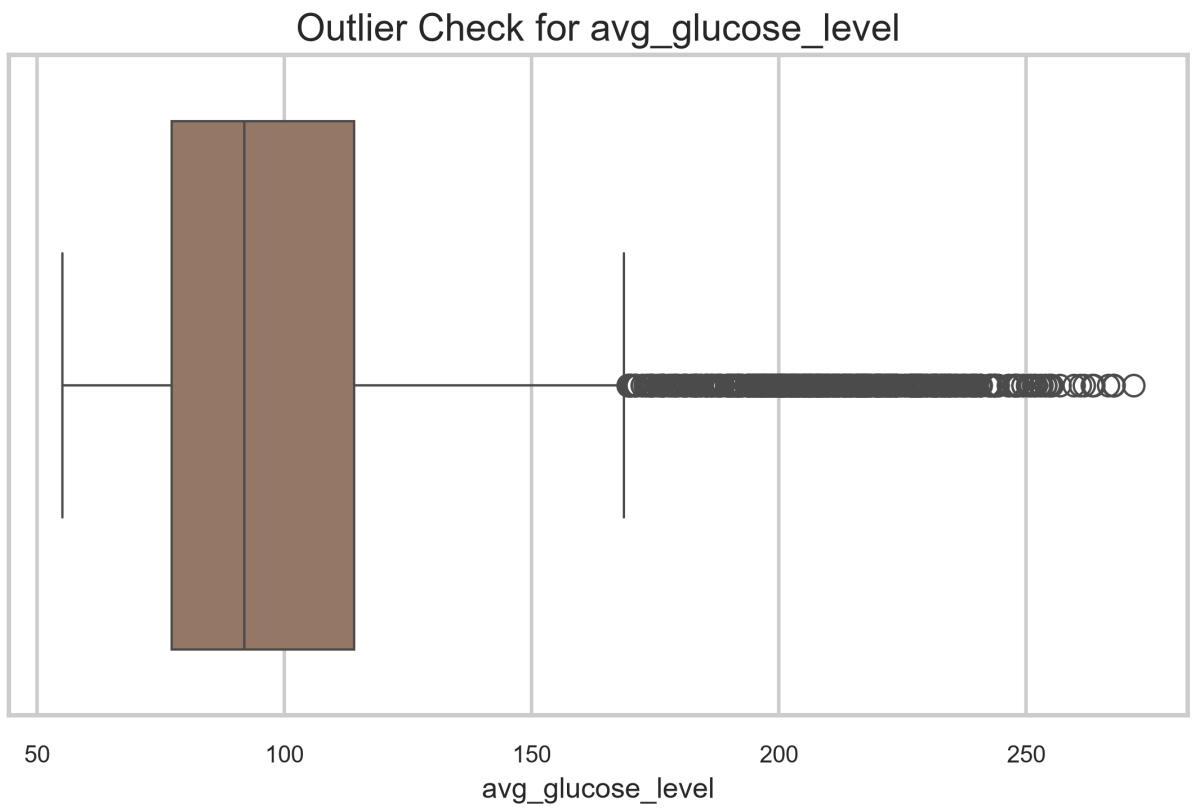


Figure 49. Glucose outlier boxplot.

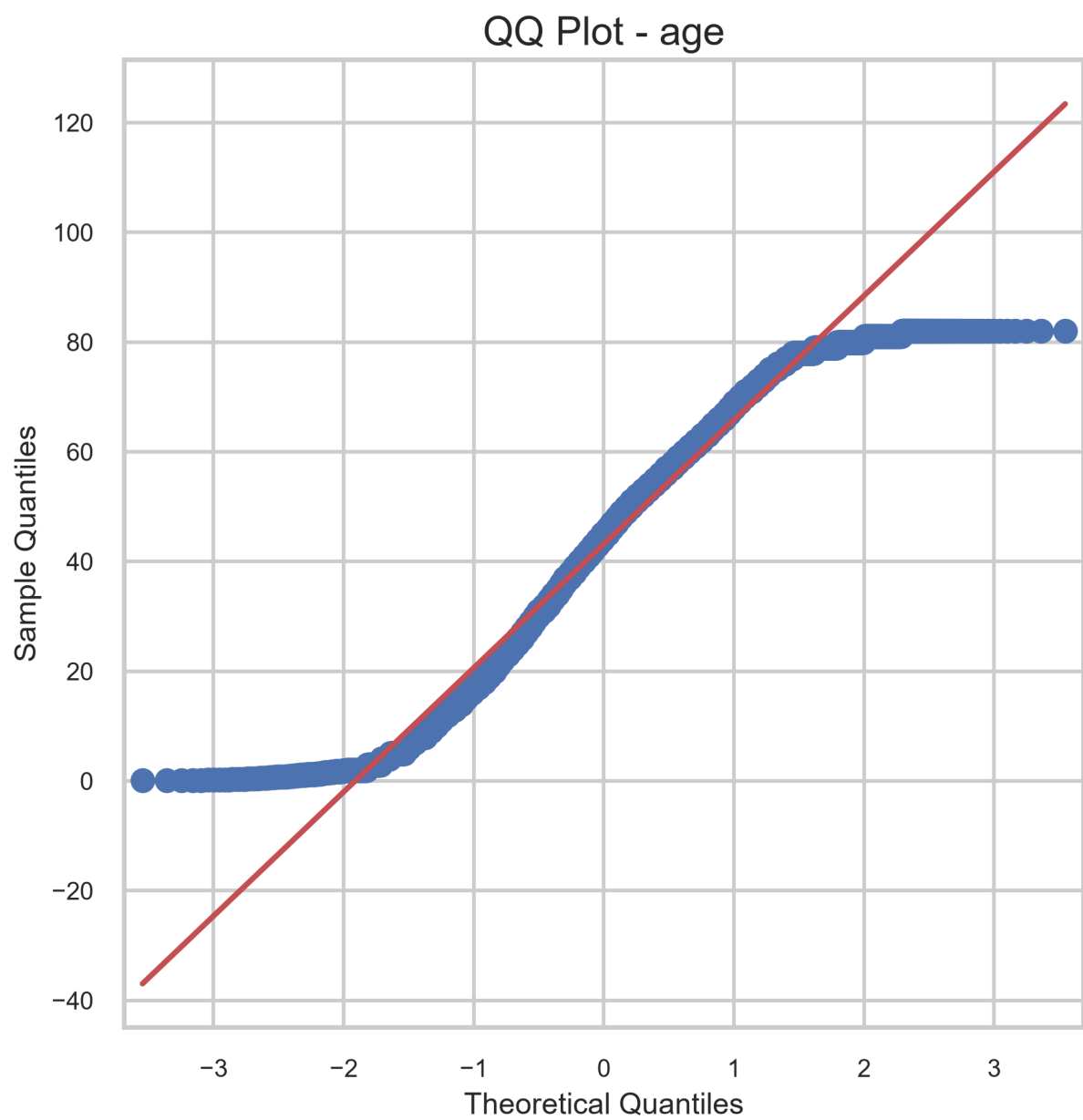


Figure 50. Age QQ plot.

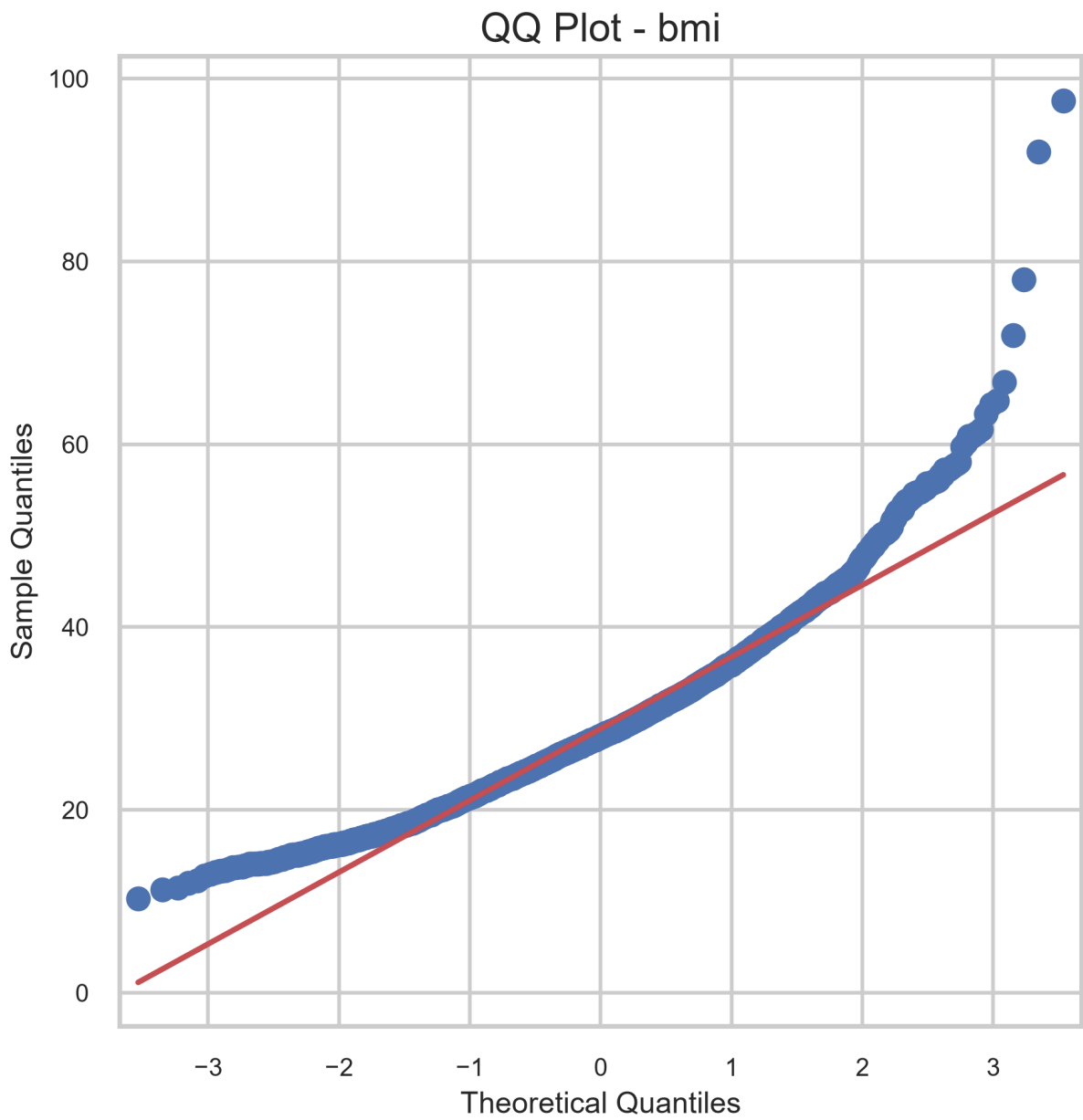


Figure 51. BMI QQ plot.

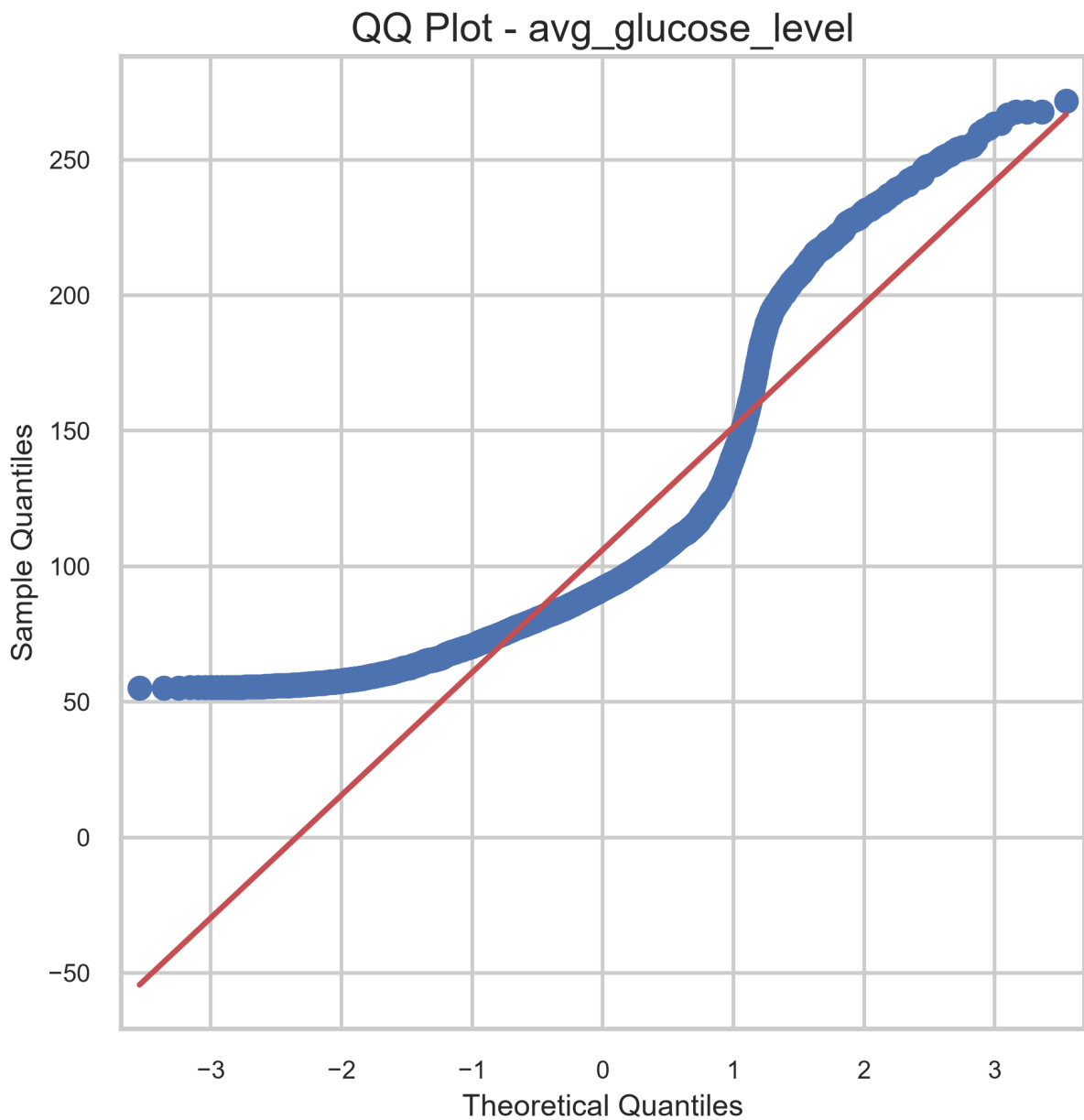


Figure 52. Glucose QQ plot.

Feature	Lower Bound	Upper Bound	Outlier Count	Outlier %
age	-29.0000	115.0000	0	0.00%
bmi	9.1000	47.5000	110	2.15%
avg_glucose_level	21.9775	169.3575	627	12.27%

Feature	Normaltest Statistic	P-value	Normal at 0.05
age	1120.5286	0.000000	False
bmi	1021.1795	0.000000	False
avg_glucose_level	1328.9358	0.000000	False

Analysis

Outlier counts under the IQR rule are 0 for age, 110 for BMI, and 627 for glucose.

Average glucose level has the clearest outlier spread.

The QQ plots confirm that all three numerical variables deviate from normality.

Normality testing marks age as False, BMI as False, and glucose as False at the 0.05 level.

Conclusion

The numerical features are non-normal and outlier-prone.

Analysis

Dataset dimensions: 5,110 rows and 12 columns.

Missing values: 201 missing BMI values.

Duplicate rows: 0.

Numerical features: age, bmi, avg_glucose_level.

Categorical features: gender, work_type, ever_married, Residence_type, smoking_status.

Target imbalance: stroke class 0 dominates at 95.13% while class 1 is only 4.87%.

Age has the strongest relationship with stroke.

Glucose and heart disease also show clear positive association with stroke.

BMI is missing for a subset of rows and is moderately skewed.

Hypertension and heart disease are important clinical signals.

Outliers are most visible in glucose, followed by BMI.

Correlations are moderate rather than extreme, which is useful for classification.

The numerical variables are not normally distributed.

Preprocessing should include imputation, encoding, and imbalance handling.

Conclusion

The stroke prediction dataset is suitable for machine learning, but preprocessing is required before modeling. The EDA shows a strongly imbalanced target, missing BMI values, right-skewed numerical variables, and clinically meaningful relationships with age, glucose level, hypertension, and heart disease. These patterns suggest that the dataset can support a classification model, but only after imputation, encoding, scaling, and imbalance-aware training are applied. The current dataset quality is good enough to proceed to the next phase, provided that the preprocessing pipeline is carefully designed.

Additional Figures

Figure A1. Missing Values Percentage.

